VDI|VDE|IT

Study

on the

National Innovation System in Syria

Final Report

VDI/VDE
Innovation + Technik GmbH

Uwe Seidel
Dr. Wolfgang Domröse
Dr. Gerd Meier zu Köcker

Steinplatz 1
10623 Berlin

December 2009

On behalf of

gtz

Federal Ministry
for Economic Cooperation
and Development

Table of Contents

1 Preface 5

1.1 Concept of National Innovation Systems 5

1.2 Scope and Objectives of the Study 7

1.3 Economic Situation in Syria 8

1.4 Acknowledgements 11

2 Methodology 13

3 Executive Summary 19

4 National Innovation System in Syria 23

4.1 Policy Level 24

4.2 Institutional Innovation Support Level 30

4.3 Programmatic Innovation Support Level 35

4.4 Innovation Capacity Level 41

5 Scope of Intervention in Syria 53

6 NIS of Syria, Jordan and Egypt - Comparative Overview 63

7 Appendix I: Programme-Descriptions and Other Related Documents 67

8 Appendix II: Literature 73

9 Appendix III: List of Interview Partners 75

1 Preface

1.1 Concept of National Innovation Systems

The competitiveness of nations and regions is nowadays not determined by single companies, but more and more by the innovative activities of entire industries and branches. For this reason, regional and national competitiveness has become the central topic for the economic and technology policy world-wide. Innovative firms grow faster and are more likely to survive during a recession. The benefits of innovation are not only limited to the original innovator though; by its own nature the production of innovation creates knowledge spill-overs that allow other firms to benefit from the initial innovation in terms of increasing total factor productivity. In turn this can create the condition for a virtuous circle of economic growth from which the whole of society can benefit.

Stimulating research and innovation can be considered as one of the major tasks of a national approach to increase the wealth of a domestic economy. It aims at different objectives, like economic growth, full employment, stable prices or positive balance of payments from international trade. Insufficient innovation was recently considered a major cause of Europe's disappointing growth performance[1]. Not surprisingly, then, stimulating the innovation performance of European member states is one of the main objectives of respective policy-makers. Nowadays, innovation is also of high priority in the Middle Eastern countries and surrounding areas. Plenty of policy measures and support schemes aimed at innovation have been implemented or are under preparation, with different impact. The diversity of these measures and schemes reflects the diversity of the framework conditions, cultural preferences and political priorities in the Middle East. Policy makers can stimulate the innovation ability of a national innovation system (NIS) by setting appropriate framework conditions as well as by public investments in infrastructure, education and funding innovation stimulating programmes.

All these measures and related effort aim to improve the performance of a national innovation system (NIS). Although there is no harmonised definition, these both options try to better explain what is meant by a NIS

- *.. the network of institutions in the public and private sectors whose activities and interactions initiate, import, modify and diffuse new technologies*[2]

- *.. the elements and relationships which interact in the production, diffusion and use of new and economically useful knowledge ... and are either located within or rooted inside the borders of a nation state*[3]*.*

[1] Expert group of the European Commission (chaired by former Prime Minister of Finland Esko Aho), Report “Creating an Innovative Europe”, 2006

[2] Freeman, C., “The National System of Innovation in Historical Perspective”, Cambridge Journal of Economics, No. 19, pp. 5–24, 1995

[3] Lundvall, B-Å. (ed.), National Innovation Systems: Towards a Theory of Innovation and Interactive Learning, Pinter, London, 1992

There is no formal common organisational framework for a NIS, neither in the Middle East nor elsewhere in the world. The main elements of a NIS in terms of education and research institutes, firms, industrial parks, incubators, governmental institution, etc. exists, but differs in terms of how they are coordinated or meshed. The key players in a national innovation concept typically include:

- Governmental and public authorities
- Firms
- Educational entities
- Research institutions
- Intermediaries
- Banks and financial institutions
- Other enablers.

Innovations typically are the result of a complex set of relationships among actors in the Innovation System, which includes all actors mentioned above. Innovations are related to new technologies, products, processes and increasingly, new service. For policy-makers, a better understanding of their own NIS can help to identify leverage points for enhancing innovative performance and overall competitiveness of a nation. It can assist in pinpointing mismatches within the system, both among institutions and in relation to government policies, which can thwart technology development and innovation. Policies, which seek to improve networking among the actors and institutions in the system and which aim at enhancing the innovative capacity of firms, particularly their ability to identify and absorb technologies, are most valuable in this context. As a consequence clusters became more and more the focus of today's innovation supporting actions[4].

The concept of NIS rests on the premise that understanding the linkages among the actors involved in innovation is a key factor to improving technology performance. Innovation and technical progress are the result of a complex set of relationships among actors producing, distributing and applying various kinds of knowledge. The innovative performance of a country depends to a large extent on how these actors relate to each other as elements of a collective system of knowledge creation and in what extend they utilise technologies. These actors are primarily private enterprises, universities and public research institutes. The linkages can take the form of joint research, personnel exchanges, cross patenting, purchase of equipment and a variety of other channels[5].

[4] Meier zu Köcker, Gerd, Clusters in Germany , 2008

[5] National Innovation Systems, OCED Report, 2005

1.2 Scope and Objectives of the Study

Although significant progress has been made in many cases, NIS in Northern Africa and the Near East are still seeking for efficiency and effectiveness to reach sustainable impacts on innovation and competitiveness. For policy makers the main challenge remains to set up appropriate measures to stimulate innovations, which are demand oriented and tailor-made with the financial investments available. Although it is well understood that stimulating innovations and improving the NIS is a task of public authorities, these still hesitate to really commit themselves and significantly invest in related measure and infrastructure. One of the main reasons is that a NIS appears to be very complex and influenced by many determinants. Thus, it is very difficult for policy makers to decide where or how to start.

The main objective of this study is to analyse the NIS of Syria according to its main determinants. It is quite clear that other studies on the NIS of Syria already exist, including recommendation for improvements. But these studies and the corresponding recommendations appear sometime very generic or too general for policy makers to turn them into practice. Voluminous reports and plenty of recommendations are the outcome, which overloads policy makers in these countries, who are not that experienced in innovation policy matters. Knowing that their NIS needs to be improved, the crucial questions for policy makers are: where and how to start, even if the capability for public investments is low, where can the Deutsche Gesellschaft für Technische Zusammenarbeit (GTZ) GmbH provide some limited input generating maximum impact? These questions will be answered by means of this study and the methodology used.

The GTZ defines a NIS to be the interaction between the four pillars of technological capabilities. These pillars are:

- The political, legal regulatory framework that stimulates learning and innovativeness;
- The education system which includes the comprehensive primary education, technology-related secondary education, vocational training, higher education and continuing education at all levels;
- Technology-related and R&D institutions, whose roles include intellectual rights protection, basic research, R&D financing, technology transfer and extension, technology assessment, technology information and dissemination, consultancy firms and testing, measuring, quality control and standards certification;
- Innovation capability at firm level, which includes the existence of creativity of employees at all levels, innovation-friendly organisational structure and networked research and development.

In short the GTZ understands that the NIS is a network between the state with its provided frame conditions, the technology-related and R&D institutions, the education system and the industry level.

The study focused on the following subjects and tasks

- To analyse the existing strategies for industrial development as well as SME and innovation policy
  - Get a better understanding of the NIS
  - Identify missing linkages/ challenges in the NIS
  - Identify missing linkages between R&D institutions and other entities
- To develop different options for the promotion of innovation and technology in consideration of existing support programmes
  - Give an overview available policy tools
  - Develop different recommendations as foundation for decision-making in terms of possible interventions and funding priorities

The study is structured as follows: The research results are documented and discussed in chapter 4; recommendations for future intervention are made in chapter 5. In addition, a comparative overview concerning the main differences between the NIS of Syria, Jordan and Egypt are defined and discussed in chapter 6. Selected examples of innovation related programmes and literature are described in the Appendix.

1.3 Economic Situation in Syria

Located at the intersection of Europe, Asia and Africa, the Syrian Arab Republic (Syria) is bordered by Lebanon to the west, Israel and Jordan to the south, Iraq to the east and Turkey to the north.

Syria can be characterised as a middle-income, developing country with a diversified economy mainly based on agriculture, industry and energy. Syrian industry is dominated by petrochemicals, although the textile sector plays an important role, too. Employment depends very much on agriculture, whereas the production of raw cotton alone accounts for 15% of employment. The Syrian economy grew by an estimated 3.3% in real terms in 2007 led by the petroleum and agricultural sectors, which together account for about one-half of GDP. Syria's GDP is forecasted by 96.53 billion or $ 4.900 per capita (2008 est.) with a growth of 2.4% (2008 est.). The main sources of the GDP are shown in figure 1[6].

[6] World Bank, 2008

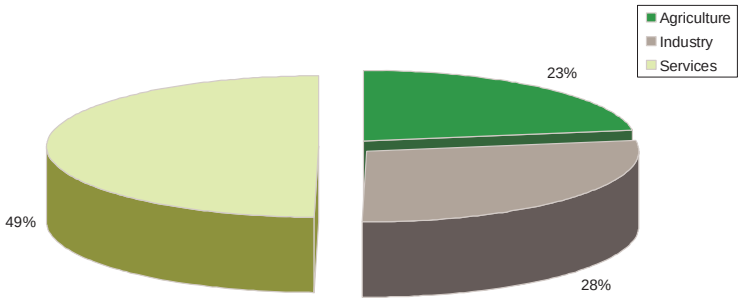


Figure 1 Origin of GDP according to sector[6]

The Global Competitiveness Report 2008-2009 draws the following picture about the situation in Syria[7]:

Syria is a still largely state-controlled economy that only recently embarked on a reform path. The successful reforms of the business environment are reflected in the relatively favourable and improving assessment of institutions, although business leaders still identify corruption and the lack of transparency of government policymaking as major shortcomings.

The government has redirected its economic development priorities from industrial expansion into the agricultural sectors in order to achieve food self-sufficiency, enhance export earnings and stem rural migration. Due to sustained capital investments, infrastructure development, subsidies of inputs and price supports, Syria has gone from a net importer of many agricultural products to an exporter of cotton, fruits, vegetables and other foodstuffs. One of the prime reasons for this turnaround has been the government's investment in huge irrigation systems in northern and northeastern Syria, which was part of a plan to increase irrigated farmland by 38% over the next decade. Syria's petroleum industry accounts for a majority of the country's export income. The government has successfully begun to work with international energy companies to develop Syria's promising natural gas reserves, both for domestic use and export[8].

Consequently, Damascus has implemented modest economic reforms in the past few years, including cutting lending interest rates, opening private banks, consolidating all of the multiple exchange rates, raising prices on some subsidised items, most notably gasoline and cement and establishing the Damascus Stock Exchange - which started operations in 2009. The state has started to privatise a number of public companies, improve the business climate and simplify cus-

[7] The Global Competitiveness Report 2008-2009, World Economic Forum, 2008

[8] Tsipouri, Elsayyard, Elsayad: Diagnostic Study of Syrian Innovation System, 2008

toms formalities. Certain state monopolies will be opened to competition, notably in metallurgy, textiles and dairy products. Special attention is also given to education and training. Ad hoc economic liberalisation continues to provide hope to Syria's private sector. In 1990, the government established an official parallel exchange rate (neighbouring country rate) to provide incentives for remittances and exports through official channels. This action improved the supply of basic commodities and contained inflation by removing risk premiums on smuggled commodities.

Nevertheless, the economy remains highly controlled by the government. Long-run economic constraints include declining oil production, high unemployment and inflation, rising budget deficits and increasing pressure on water supplies caused by heavy use in agriculture, rapid population growth, industrial expansion and water pollution. The government has begun to address structural deficiencies in the economy such as the lack of a modern financial sector through changes to the legal and regulatory environment. In 2001, Syria legalised private banking. In 2004, four private banks began operations. Beyond the financial sector, the Syrian government has enacted major changes to rental and tax laws and is reportedly considering similar changes to the commercial code and to other laws, which impact property rights. Increased attention has also been given to innovation and competitiveness. Policy makers on national and regional level have understood that increased innovation capability can be considered as a key for improved competitiveness of the nation. Some actions have been undertaken to address this issue, much more efforts will be needed.

Foreign aid to Syria is still low compared to some neighbour countries, especially in the high tech sector (figure 2). The World Bank reported that as of July 2004 it had committed a total of US$661 million for 20 operations in Syria.

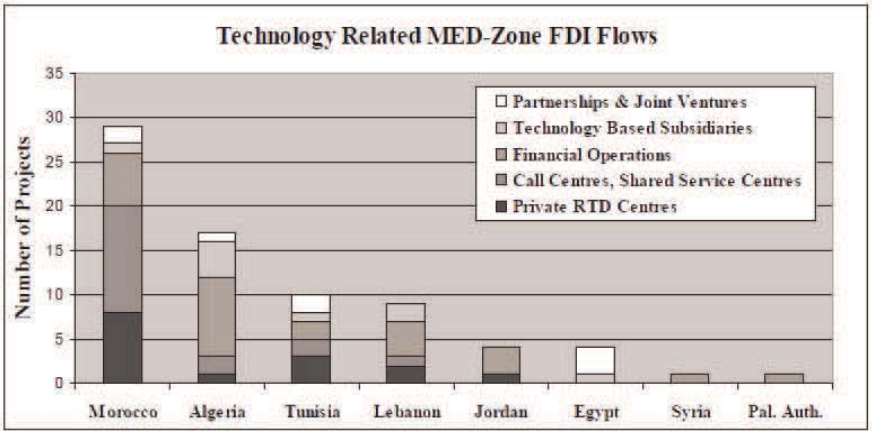


Figure 2 Foreign direct investment related to technology[9][10]

[9] Based on data provided in ANIMA Papers and Reports No. 9 'La Méditerrannée Intelligente Innovation, Pôles Technologiques et Attraction de l'Investissement, 2006

[10] EUROPEAN TRENDCHART on INNOVATION, MEDA-Region, 2006

There are quite a few important achievements, which, combined with the positive climate in the energy exports, may pave the way for succeeding in the modernisation process. The government has recently moved towards *deregulated state-owned companies*. Recent laws grant state owned enterprises more managerial flexibility and autonomy and introduce principles of corporate governance. Another important element of all changes is the decision to open up in the international market and hence the need to improve competitiveness. Over the past years, the government has made trade liberalisation a key element of its reform agenda, along with Syria's participation in the Great Arab Free Trade Area and the negotiation of an Association Agreement with the EU. Moreover, Syria recently renewed its application for membership of the World Trade Organisation (WTO). Syria has developed new markets in recent years and engaged in bilateral preferential agreements with a more diverse group of economic partners including Turkey, Russia, China, Iran, India and Ukraine. Geographical patterns of trade flows have shown a larger integration with Arab countries as a result of the full implementation of the GAFTA. These countries represent the primary market for non-oil Syrian exports (35% of total Syrian exports), while accounting for close to 17% of total Syrian imports (Saudi Arabia, Egypt and UAE being the main partners). The Asian countries have considerably strengthened their position as suppliers to the Syrian market, representing around 28% of total Syrian imports.

Syria signed the *Euro-Mediterranean Charter for Enterprise* in October 2004 and undertook to give priority to simple procedures for enterprises, education in entrepreneurship, and easier access to finance and better market access. This commitment has materialised through the integration into the Five-Year Plan 2006-2010 of the Charter principles, which provide the government with guidelines for pro-SME and private-sector development policies. However, the opening up is a double-edged sword: whether there is no doubt that there will be considerable import penetration, as soon as the liberalisation takes place, it is not equally clear that exports will grow as rapidly: unless Syrian enterprises succeed in becoming more productive/innovative and meet global quality/price standards, there is a clear danger that imports will displace employment and trigger a vicious circle until productivity rises.

1.4 Acknowledgements

The study was carried out between August and September 2009 by VDI/VDE-IT in close cooperation with GTZ and local partners.

One stay in Syria by VDI/VDE-IT experts served the purpose of conducting a series of interviews with actors within both innovation systems, consisting mainly of representatives of ministries, representatives of universities, technology centres, innovation support programmes and stakeholder organisations from industry. The complete list of interviewees is given in Appendix III.

We are very grateful to the GTZ-colleagues in Syria, who strongly supported our work. We would also thank our network partners in these countries for their assistance.

Berlin, October 2009

2 Methodology

Theoretic models and descriptions of NIS as well as their analyses have been increasing in number since the early 2000s. Assessing a country's innovation system is a challenging task because of the enormous number and variety of influences on national innovation capability. For years, economists have tried to understand what determines the performance of the competitiveness of nations, the outcomes of innovation approaches and the wealth of nations. As a consequence, many reports and analyses of NIS have been generated. Although these reports were excellent, described the main elements of a NIS and caught its main strengths and weaknesses, the benefits in terms of usable results were mostly limited. The main reason is that the applied methodologies of the studies were not perfectly linked with the circumstance how policy makers think and operate. Recommendations are neither prioritised nor ranked according to their complexity when turning into practice.

Policy makers, especially in emerging and developing countries, usually are looking for well structured descriptions of a NIS and clear recommendations how to improve the functionality of a NIS and what kind of specific measures were recommended. They do not ask for receiving scientific models of the functionality of a NIS or how the single actors are linked. As far as embryonic or not well established NIS are analysed, they have mainly been compared with those that are matured. The consequences are plenty of weaknesses found and recommendations made. Often policy makers are confused rather than getting a clear guideline on how to start corrective actions. Such reports have often failed to provide clear information or recommendations how to start and how to gain a high leverage effect (especially when public investments were limited). When it comes to GTZ concerns, it is of main interest to identify those determinants of a national innovation system, which can be improved by the tools and financial means available for GTZ.

The present report falls into the new tradition of indicator-based studies based on quantitative data generated by the evaluation of expert interviews. The Global Competitiveness Report or the European Scoreboard on Innovation is an excellent approach to benchmarking the performance of NIS. The latter one is mainly intended for well matured economies and NIS and cannot sufficiently be applied for developing or emerging countries since the statistical base is often insufficient. The Global Competitiveness Report uses a mix of statistical data and expert interviews, but it is focussing on the competitiveness of nations.

The overall approach we have applied in this study can be divided in the following steps:

- Analysis of literature on NIS
- Conducting interviews with experts on NIS
- Evaluation and measuring the results
- Identifying those determinants which appear to high impact with less efforts
- Deducting recommendation how to improve the prioritised determinants

The study captures this challenge by providing an indicator based assessment of many different determinants, each of which reflects one aspect of the complex reality that we call innovation. We

grouped all these determinants according to the macro, meso and micro level figure 3 describes the different dimensions and its actors.

Level		Actors	Functionality within a NIS
Macro	Policy	Public authorities, policy makers	Governing and setting framework conditions of a 'NIS
Meso	Institutional innovation support Programmatic innovation support	Institutional innovation supporter or public funded initiatives / programmes	Institutions and initiatives are tools to turn innovation policies into practice
Micro	Innovation capacity	Firms, academia, educational institutions, etc.	Main beneficiaries of support measures and main producers of knowledge, innovation, technologies, products

Figure 3 Levels and actors within a NIS

Macro Level – Innovation Policy Level

On a macro dimension national and regional innovation policies directly influence the framework conditions of a NIS. Laws, decrees and regulations, etc. on that level are often path breaking in a positive or negative sense. The readiness of public investments in innovation is a direct outcome on decisions made on policy level. But, decision on this level can only influence the framework condition of a NIS and do not turn innovation into practice.

Meso Level - Institutional Innovation Support Level

Institutions operating on this level are typically technology transfer centres, clusters, innovation service providers and funding agencies. They can be considered as the tools to turn political decision on innovation into practice. In emerging countries, such institutions are mainly publicly owned. Their main tasks are to serve the stakeholders in order to increase their competitiveness and capability to innovate. Institutions on this level generally do not own separate programmes to fund innovation support measures rather than providing in-kind contributions by training, consultation, conducting applied R&D or product improvements. Nevertheless those institutions are one of the major pillars to improve the innovation capabilities of firms, especially in those countries where public investments are limited.

Meso Level: Programmatic Innovation Support Level

By programmatic innovation support we mean public funding programmes and initiatives used to turn innovation policy into practice. It represents the second important pillar to improve the innovation capabilities of the stakeholders within a NIS. Such programmes can be managed by policy makers or innovation support institutions. Nevertheless, measures on that level need significant public investments.

Micro Level: Innovation Capacity Level

The micro level builds the umbrella for the main practical actors and enablers within a NIS, like SME, entrepreneurs, universities, public or private R&D institutions, innovators or financial facilities.

The different dimensions are influenced by certain determinants. These determinants are of dedicated interest for our analyses since they can be influenced and improved by appropriate measures. In summary, the pattern of determinants on all four levels mostly affects a NIS. Although we use the four levels separately, we acknowledge that there are plenty of interdependencies and links between them. But it seems to be appropriate to consider the levels separately during the analyses of NIS in Syria. We are also aware that the different determinants can influence a NIS differently. The ideal way for Syria to improve the outcomes of its NIS is not the same as the best way, for example in Italy. Thus, we did not evaluate the pillar or the respective determinants, although it is obvious that certain determinants are usually of higher relevance than others. At this point we want to point out that a NIS is, of course, also affected by factors coming from outside the country. Within a globalised world, all NIS are affected by external influences. But, since these factors cannot be controlled by national policy makers and actors within a NIS, they will not be considered in the present analysis.

In total we identified a core set of 29 determinants below the three levels, which, by our perspective, are considered worth to be analysed within this study. All of them directly influence the efficiency of a NIS. All determinants can directly be addressed by different measures. Some of them by short-term and with low efforts, others need long periods of time for improvements, combined with significant investments. Improving a certain determinant can have manifolds positive impacts.

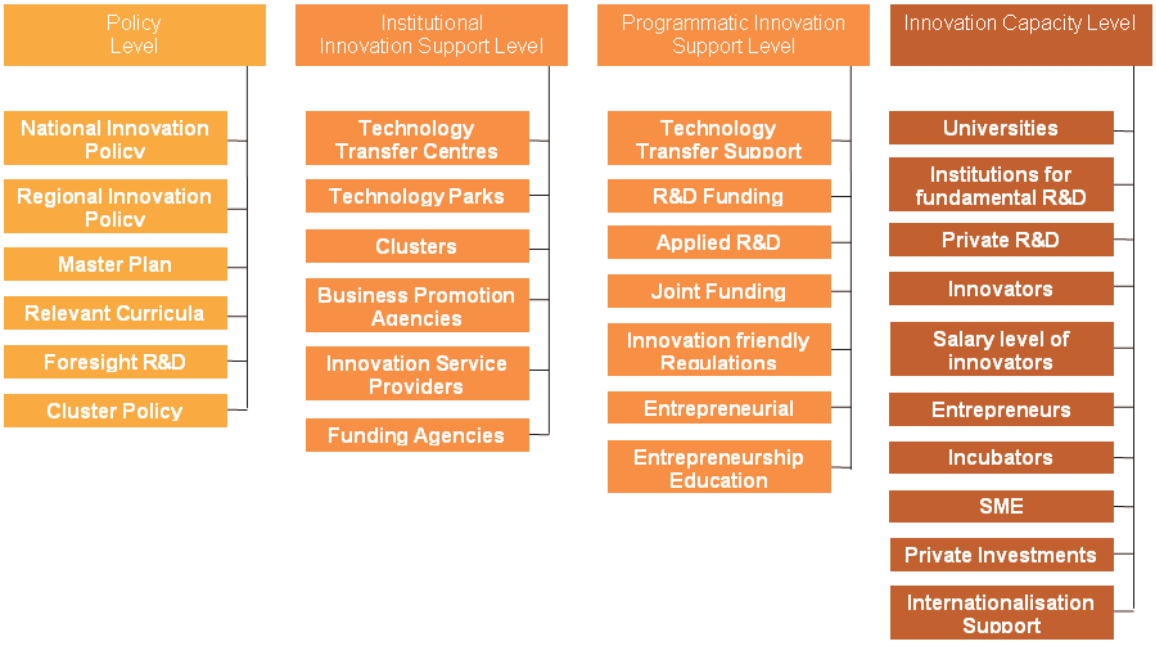


Figure 4 Main determinants of a national innovation system

In order to assess the stage of development of all determinants, we have designed between three and five questions for characterising the 29 determinants accordingly. The onsite assessments were done by national experts as well as by the expert team of VDI/VDE-IT as explained below.

Expert Opinion Survey (EOS)

The model used draws on a wide range of data from the Expert Opinion Survey (EOS). The EOS meets the need for up-to-date and far-reaching data, providing valuable qualitatively information for which hard data sources are scarce or nonexistent. The survey was completed by at least 20 national experts for each country. We asked the experts to provide their opinions on various aspects of innovation and the innovation environment in which they operate. The data gathered thus provide a unique source of insight and a qualitative portrait of each nation's innovation concept and how it compares with the situation in the other country.

The questions in the study follow a structure that asks the interviewees to evaluate, on a scale of 1 to 4, the current conditions of their particular operating innovation environment. At one end of the scale, 1 represents the worst possible operating condition or situation and at the other end of the scale, 4 represents the best (see figure 5).



Figure 5 Example for a typical question

The experts for the EOS have jointly been selected by GTZ as well as by VDI/VDE-IT and its local partners in order to cover the complete spectrum of determinants of a national innovation system. This valuable collaboration helped to ensure that the interviews were conducted in a consistent manner across both countries. In addition, our partners supported us in explaining the approach at the national level. Nearly all of these experts have been interviewed by VDI/VDE-IT with support of local partners and most of them have completed the survey after the interview. Those experts, that have not been available during the stay of VDI/VDE-IT in the specific country, have been asked to complete the survey on their own.

The Indicator Approach

Based on the findings of the EOS, we then were in the position to calculate appropriate indicators based on the evaluation of the questions grouped around the one determinant. We also used a scale which was very similar and therefore comparable. When it come to the scale,

- Indicator of “1” represents the determinant at its worst operating condition or situation pointing out that a certain determinant is very poorly developed or not existing. This value was generated if all the questions belonging to that determinant were answered by “1”, too.
- Indicator of “2” means, that a certain determinant basically exists and has shown first positive impacts. Nevertheless, there is a strong need to improve the efficiency or functionality.
- Indicator of “3” means, that a certain determinant is matured and has shown positive impact on the performance of a national innovation system over a longer period of time. Nevertheless, there is still room for further improvements towards an excellent performance.
- Indicator of “4” represents the determinant at its best operating condition. It is strongly developed and well operating in practice over a long period. However, some very slight improvements may be still possible

Indicator values above 3 are characteristic for well developed industrial countries where all determinants are established and well functioning, some slightly better than the others. A value between 1.5 and 3 means the determinant already exists and is in the phase of development. Values below 1.5 mean that a specific determinant may exist, but is not really operational in practice or it needs improvement. This is characteristic for those countries that have a quite weak NIS in force.

The following figure gives an example for the chosen approach to visualise the results out of the expert interviews. The green tab marks the individual determined results.

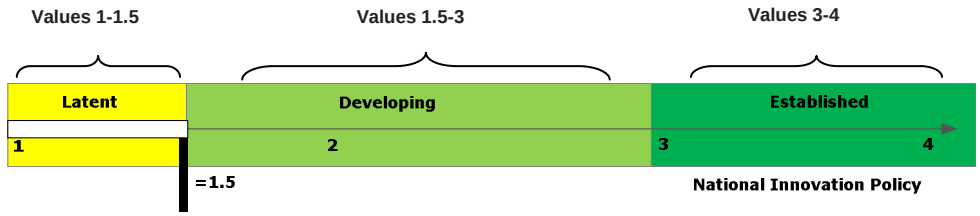


Figure 6 Example for the visualised results out of interviews

Analysis of Potential Improvement

Describing the determinants is important in order to better understand the quality and stage of development of a NIS. If a NIS should be improved, certain determinants must be addressed. It is quite clear that certain determinants are easy to improve, while others are much more complex. For that reason, we grouped the determinants according to their complexity for improvement. We used a portfolio with two different scales. One scale represents the “efforts needed” (how extensive is the needed amount of investment to enhance the performance of the determinant?), the other represents the “expected impact” (what range of improvement can be expected?).

Thus we focused our recommendations on those which do not need a huge amount of effort, in terms of money, training, etc., but have a significant impact on the overall performance of a NIS.

3 Executive Summary

Innovation is the ability to take new ideas and translate them into commercial outcomes by using new processes, products or services in a way that is better and faster than the competition[11]. Certainly innovation cannot be organised by decree. It needs stakeholders and people being active within a NIS and only these people - scientists, researchers, entrepreneurs and their employees, firms, investors, consumers and public authorities — can make a country or region more innovative and competitive. But they do not act in a vacuum. They act with a mindset and in a framework which either discourages or incites them to enter unknown territories, the so called national innovation system.

The main objective of this study was to analyse the Syrian NIS as well as to identify future supporting measures in order to increase the competitiveness of local SME as well as the innovation capability of Syrian actors. Special focus was given to a set of recommendations for improving actions, which may be implemented and funded by external donors, like GTZ in the near future (chapter 5).

As far as the analysis of the NIS was concerned, an indicator-based approach was chosen, involving many local experts, who provided GTZ and VDI/VDE-IT with their expert opinions. The so called **ANIS**[12] approach is based on 29 determinants, which, on one side, are very characteristic for the different facets and the level of maturity of the Syrian NIS and, on the other side, are very much influencing the effectiveness and efficiency of an innovation system. This approach allows to clearly presenting the strengths and weaknesses of the Syrian NIS in a very transparent way. The determinants were calculated based on interviews with many local experts (Expert Opinion Survey), which then were calculated and transferred on a scale between “1” (not existing or in a very early stage of development) and “4” (well matured). Similar analyses, based on the ANIS approach, were also conducted in different neighbour countries, which allowed to benchmark the maturity of the NIS and its determinants of Syria, Jordan and Egypt (chapter 6).

[11] ‘Creating a National Innovation Framework’, Science Progress, Richard Nedis & Ethan Byler, April 2009.

[12] ANIS – Indicator-based **A**nalysis of **N**ational **I**nnovation **S**ystems, developed by VDI/VDE-IT

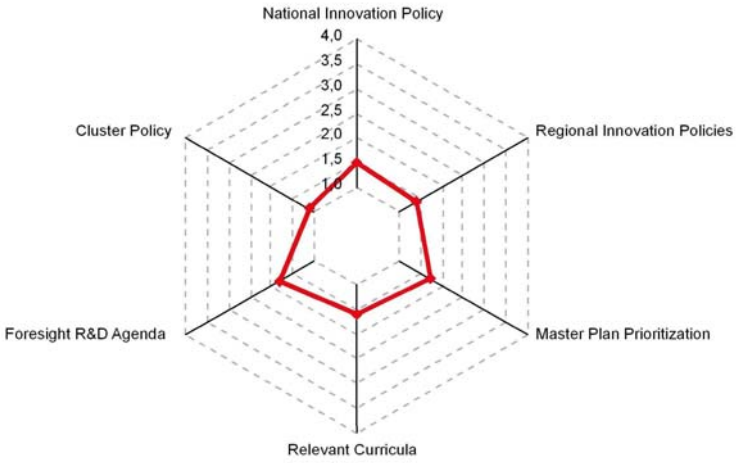


Figure 7 Pattern of the determinants on policy level

There are many evidences on all different levels that the awareness on the importance of innovation considerably increased over the last years in Syria. Pretty much progress was made in this respect. As far as the six determinants related to the **policy level** (macro level) in Syria are concerned (figure 7), most of them are in the so called developing phase. A dedicated national innovation policy does not exist as it is known in fully industrialized countries. Also on regional level such policies are missing. Nevertheless the findings reveal that elements of a national innovation policy are scattered over certain other policies, laws, regulations or are part of the actual five-year-planning. As in other national economies, similar to Syria, missing links between curricula and innovation topics, led to the situation that curricula in Syria are not sufficiently designed towards the demand of the industry and that there are very few innovation related elements implemented in the curricula of public universities. Although foresight-processes concerning the priorization of future technological domains, incl. R&D, are not officially politically supported, many experts from academia and industry in Syria seem to be involved in such discussions about future technological developments and challenges, whereas these determinants was ranked quite high compared to others.

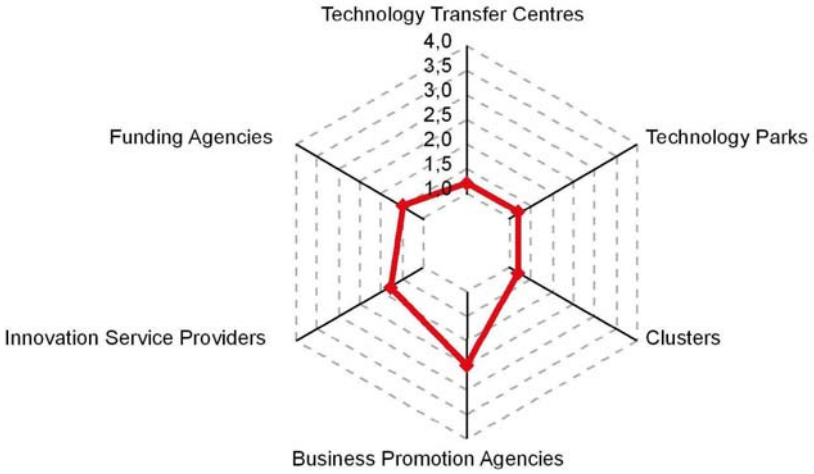


Figure 8 Pattern of the national institutional innovation support level

Assessing the **institutional innovation support level** (meso level) in Syria, the general pattern of six determinants reveals some very interesting findings. Business promotion agencies are highly recognised as tools to support firms in certain innovation matters and seem to play a comparable strong role in Syria. Also innovation service providers are operating on a comparable high level compared to the other institutional innovation supporters (figure 8). Since the Syrian government, partly together with external donors, focussed supporting business promotion agencies and certain innovation service providers in the past years, the positive outcomes nowadays become visible, resulting in a comparable high rating of the corresponding determinants. Other institutions for innovation support, like technology parks, technology transfer centres and clusters are not that far developed.

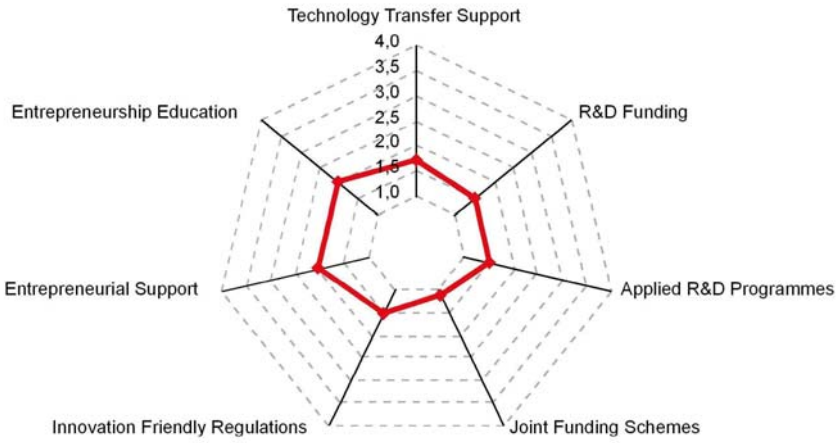


Figure 9 Pattern of the national programmatic innovation support level

The pattern of the seven determinants related to the **programmatic innovation support level** (meso level) yields some interesting characteristics as well (figure 9). Determinants related to education and entrepreneurial support are rated fairly high in Syria. There is a comparable high number of programmes, initiatives as well as institutions supporting entrepreneurs in many areas (projects like SHABAB or MASSAR as well as the Syrian Young Entrepreneurs Association, The Syria Trust for Development, FIRDOS, etc.).

Public funds for R&D seem to be principally available in Syria (for example proved by the Higher Commission for Scientific Research). But the barriers to get access to such funds are considered to be quite high in practice as well as the demand seems to be much higher than the currently available amount of funds. Thus, the corresponding determinants have been ranked quite low.

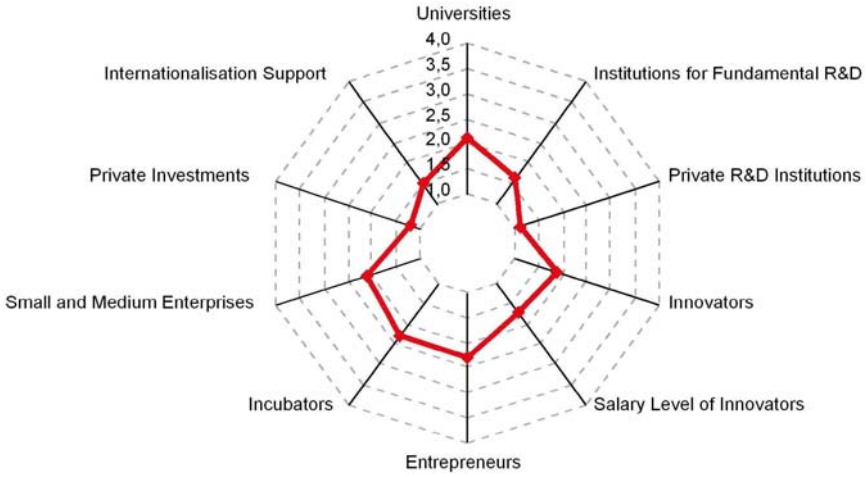


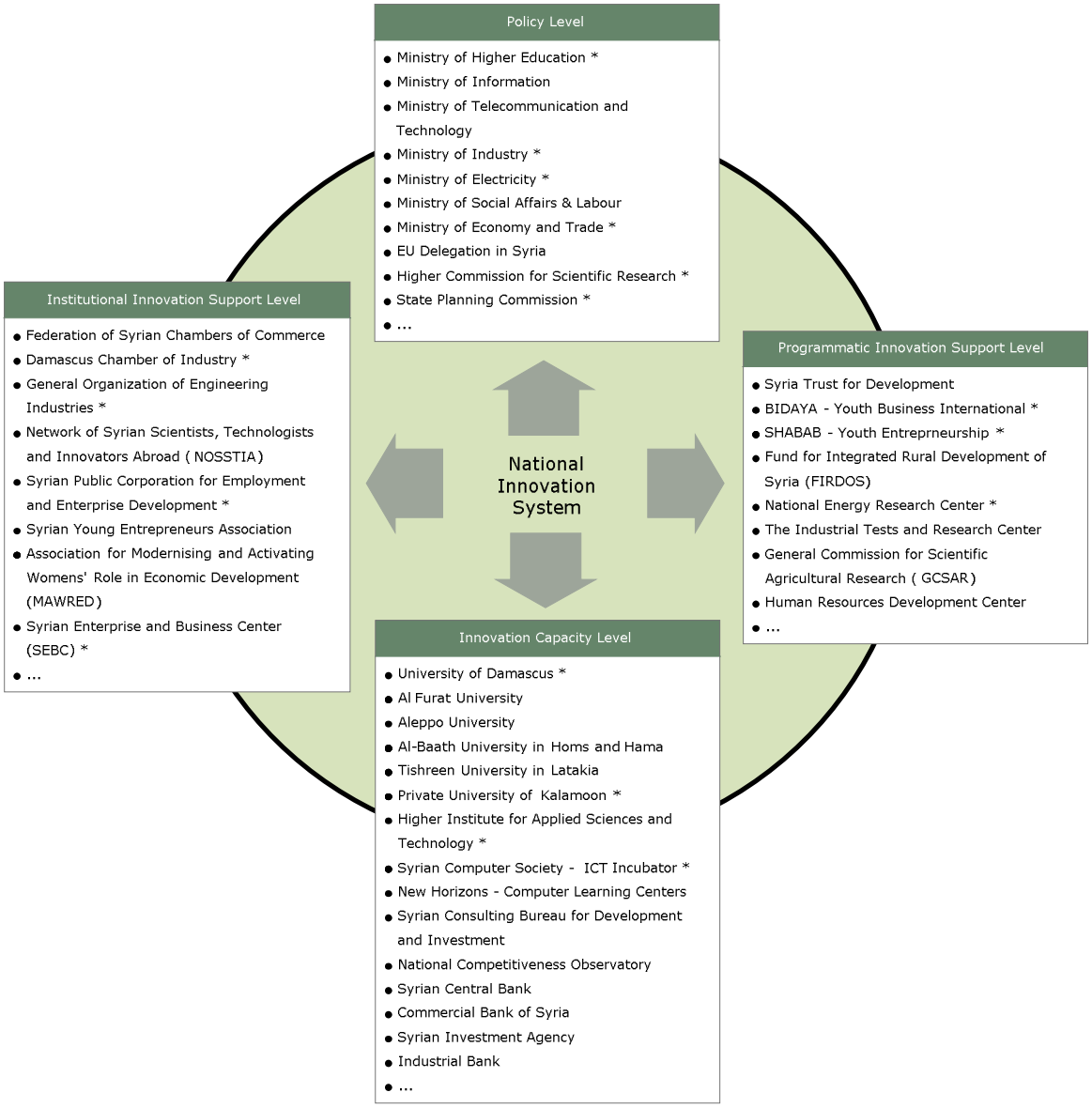
Figure 10 Pattern of the innovation capacity level

Universities are the determinant that is best developed on the **innovation capacity level** in Syria according to the interviewees (figure 10), but with conservative designed curricula. To enhance the capabilities of researchers to transfer academic results into products and services, there are promising approaches existing. Entrepreneurs became an important component for the development of an innovation culture in Syria. Most SMEs are currently not in a position to act as drivers for innovation. There are only very few innovative SMEs, which can be considered as high potentials. Private investments in R&D are still an exception in Syria.

Based on the findings a set of recommendations were identified, how to improve the effectiveness and efficiency of the Syrian NIS. It is obvious that some determinants of a NIS are easy to improve, others are much more complex. Based on a portfolio analysis especially those determinants were identified and addressed by recommendations, which can, on one hand, be implemented with reasonable efforts and, therefore, are of dedicated interest for third party donors. On the other hand, a successful implementation of these recommendations may have a significant positive impact on the Syrian NIS in the near future. These recommendations focus on

- Setting up and implementing a National Innovation Strategy involving all relevant actors
- Further improvements of the framework conditions for entrepreneurs
- Increasing the capacities of institutional innovation supporters
- Defining and implementing a national policy for setting up regional innovation networks (clusters)

4 National Innovation System in Syria



* interviewed partners

Figure 11 The main players in the Syrian innovation system (besides of the industrial actors)

If we understand a national innovation system as the constitution and interaction of different public bodies, organisations, commercial companies or supporting institutions Syria of course has a NIS. The main actors and programmes on all three different levels are shown in figure 11. Even the listed number and names account for many activities to this respect, though most of the interview partners were concerned because of the “missing glue”, a coordinated approach which is missing in many cases. Further implementation steps to address the systemic nature of innovation and to harness resources, residing in different ministries, is seen as a coherent way.

In the following chapters we assessed the 29 determinants, grouped according to the three levels of a NIS. This has been done based on the Expert Opinion Survey, which involved a significant number of the expert Interviews, as well as on the impression of the expert team. This approach led to a whole picture of the NIS Syria. In the beginning, we grouped the determinants according to the respective levels (macro, meso and micro) and gathered them in one spider graph. Then we described each single determinant, starting with a short summary, followed by a more detailed description and a short comment, whether potential improvement may by reasonable taking the limited resources especially for third party donors into account.

4.1 Policy Level

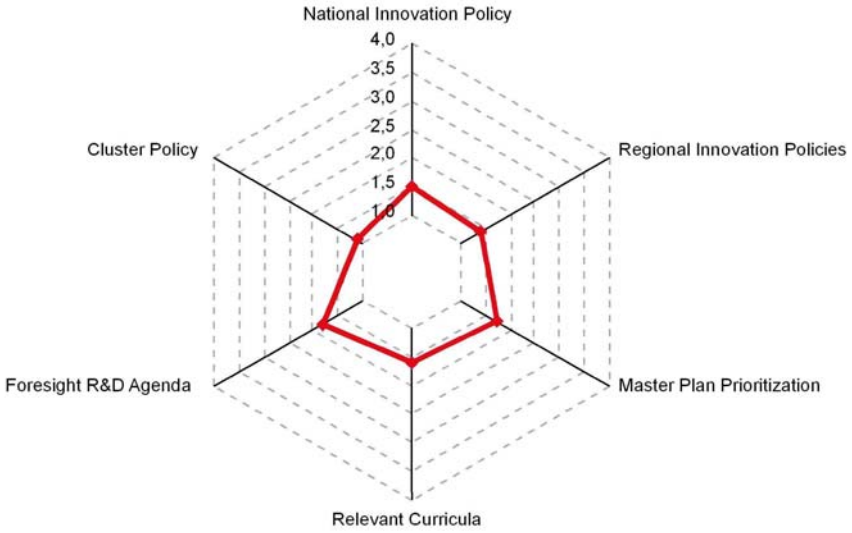
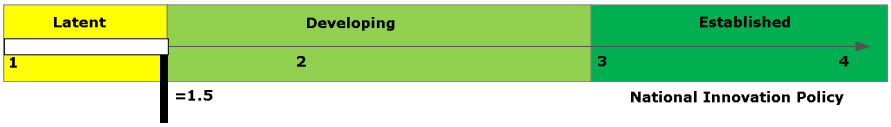


Figure 12 Indicator analysis policy level - Syria

Figure 12 shows a clear picture of the values across the six determinants of the policy level. The pattern clearly shows which determinants of innovation policy are further developed than others. Strategy-processes to develop innovation related policies are still in a very early phase in Syria. Most of the scores are on the boarder from latent process of defining to the beginning of the developing phase. Innovation is a strong topic on the policy agenda, but still more related to R&D; cluster development is not in the focus so far.

In the following the six determinants are assessed and described in more detail.

Findings: National Innovation Policy



During the interviews with relevant experts at all levels, the existing mix of policies, laws, measures and institutions were mentioned very often. There is not a written national innovation policy available, but innovation topics are part of the actual five-year-planning.

Syria is still in the process of reinvention. The former centrally planned economy is on the way to change into a market oriented and more privatisation fostering system. These circumstances were mentioned in nearly all interviews as the most positive development during the last years. According to the EOS, Syria has no dedicated national innovation strategy in force. There are many appropriate partners in all level available (s. figure 12), but they act without efficient coordination. It is very obvious and mentioned by most of the interview partners that more coordination between measures and actors is urgently needed to enhance efficiency.

The different elements of the NIS are fitting insufficiently together or even don't know about each other. SME are some of the most important elements in any NIS, they are responsible for the transfer of knowledge into revenue and therefore they are most important for the reinvention process. The interviews and analyses stabilised the impression that the business sector is still far away from a practical involvement of demand oriented policy-making. Most of the companies are very limited concerning innovation and future-oriented R&D. At the moment there is no approach available to overcome the gap between science and industry in terms of mutual co-operation to produce new products and technologies.

The five-year-planning (differently from the 10-year-plan) in Syria is a fairly new approach of the State Planning Commission (SPC). In the past, the situation was characterised by a very low coordination of targets, activities and measures between ministries. Together with other relevant institutions (for example the Higher Commission for Scientific Research), SPC is now acting under the umbrella of the Prime Ministry and is responsible particularly for the coordination of the implementation of the Five-Year-Plan. Since the latest version of the plan some new chapters were included, with high importance for innovation development: science, technology and research & development sector. The plan is very comprehensive (over 800 pages) and is characterised by a high level of openness and self-criticism. The main objectives of the plan are the opening of public economic sectors to private investment, the upgrading of social structures and living standards and anti-corruption measures.

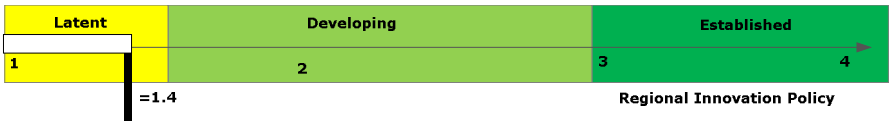
Although the interviewed experts acknowledged that there are positive developments in the strategic processes to support innovation, the implementation is still the major challenge for the future. The policy-concept is not yet fully practice-oriented, measures and activities are mainly individual cases and institutional tasks are not clearly assigned to the demand. To overcome this challenge, the Higher Commission for Scientific Research (HCSR) was established in 2005 and which acts directly under the Prime Minister as an umbrella for all research related institutions.

The main tasks of HCSR:

- Supporting the national policy for R&D as an important element of the national innovation system
- Coordination of all R&D institutions (private and public)
- Coordination of the relation between academia and industry
- Financial support of research projects (actual 20 running projects)

The process to better implement a national strategy towards innovation as a national innovation strategy for Syria seems to be implementable with reasonable effort, but comparable high impact. Most of the necessary partners expressed their serious interest to play an active role and to be involved in the strategic process. High interest was also shown in terms of implementing such a national strategy. International support could assist the responsible bodies and actors with good-practice-examples, training and hands-on consultancy.

Findings: Regional Innovation Policy



Besides of the National Plan, each region in Syria has individual economic strategy approaches, which are not really policies and most likely rather short-term oriented. According to the local professionals, these approaches are designed to the specific demands and under the regional circumstances. The findings suggest that so far there are no dedicated regional policies towards innovation.

The introduction to the Five-Year-Plan mentions the following statement: “The common feature among all local plans at the governorates, however, lies in commitment to the national planning rules, systems and methods of project selection, as well as the evaluation and follow-up systems. ...This will be implemented by the ministries and governorates in coordination with the State Planning Commission (SPC)”[13]. In result, the national guidelines for relatively new topics, like innovation, cluster and entrepreneurship will be the course of orientation for the region. They will follow good examples and successful approaches and adopt them for the regional demand.

Special treatments for less developed regions like the eastern parts or the north with a weak textile industry will be set up by the national bodies or in cooperation with international partners, like in 2005-2006, the European Development NGOs thematic programme co-financed two projects implemented by European NGOs in partnership or association with Syrian NGOs, supporting women’s micro and small enterprises in the coastal midlands area as well as strengthening the capacities of the Syrian Family Planning Association.[14]

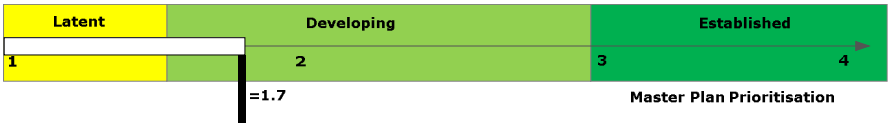
[13] Syrian Arab Republic – State Planning Commission, The Five Year Plan Approach, page 11

[14] Source:

www.delsyr.ec.europa.eu/en/eu_and_syria_new/projects2009/Projected_co_funded_by_European_Commission_through_Syrian_NGOs.htm

Regional innovation policies and appropriate actions should be based on deeper analyses of the strengths, potential and demands of the regions in terms of innovation. To assess the potential and the perspectives of different regions in Syria a deeper, well structured analysis is mandatory. Each region requires a tailor-made approach, designed to the real needs of the partner who could be responsible for the implementation. The expected impact of regional activities in the mid-term timeframe is not very high as long as a national innovation strategy is missing.

Findings: Master Plan Prioritisation

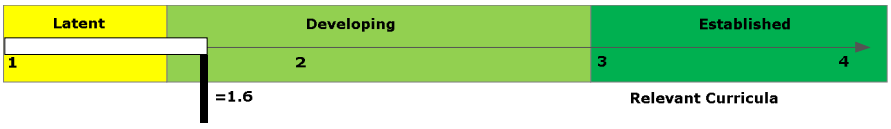


The dedicated prioritisation of certain topics related to implement innovation supporting actions is, according to the interviewees, in the developing phase. The value for this determinant is higher than of the others. Most of the interviewees mentioned the visible development in quality of the planning procedure, although practical implementation is still missing.

The Five-Year-Plan approach is sector-oriented, including branches of industry (like manufacturing or energy) and measure-oriented, including chapters about education and scientific research or science, technology, research and development. This seems to be some kind of prioritization, but so far without visible effects for the interviewed stakeholders. Many interview partners pointed out, that the definition of priorities in the Five- and Ten-Year-Plans is still very theoretic and the setting of priorities is almost realised without practicable action plans.

Same as the general planning process, the prioritisation of future oriented innovation supporting measures and their implementation could benefit from better coordinated internal procedures. Once a national innovation strategy exists, the implementation can be done easily by means of master planning. Master planning in this respect means to turn policy measures and goals into practice by certain measures and actions. This could be a field of future external support, since the efforts needed are quite low. But, the key is the availability of a Syrian national innovation strategy.

Findings: Relevant Curricula



In most of the interviews the missing link between curricula and innovation topics, as well as the missing links to the demands of the industry, was considered to be a strong issue. It seems that curricula in Syria are not designed towards the demand of the industry and that there are very few innovation related parts in the curricula of public universities.

The responsibility for the national university curricula lies with the Ministry of Higher Education. The goals of the ministry are very ambitious; it plans to develop high-quality curricula and educational programmes to increase higher education opportunities and to create an enabling environment for higher education and scientific research.

In result of interviews and own research activities, the curricula for public universities seems to some extent be far away from the needs of the local industry and as well far away to enhance innovation capabilities.

There are schools for vocational training related to the dual system with advanced equipment under the umbrella of the Ministry of Education. In all curricula, starting in 2010, entrepreneurial education will be a main part[15].

Two examples of related training measures beside the academic context:

The project SKILLS (Superior Knowledge by Intensive Labour Learning Schemes) is a joint approach of the Syrian Enterprise Business Centre and the European Training Foundation (ETF). Within the measure, selected students or young people can increase their employability by providing new skills and competences to start working in the international framework of the global economy. It aims to provide Syrian businesses with skilled employees who can help to contribute to economic development. SKILLS is focussing on entrepreneurship development and entrepreneurial learning by changing the mindset of young people and providing managerial and business related skills.

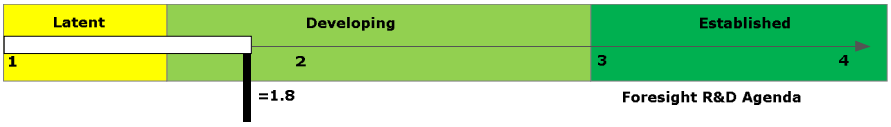
The Human Resources Development Centre (HRDC) is an institution, formed in 2005 initiated by the Chamber of Industry to create a modular training concept for an industry-related and user-oriented approach. Target groups are engineers out of the industry, who receive a very compressed and practical training, related to their daily demands. A train-the-trainer-concept to teach professionals how to teach engineers is in the status of a draft.

There is no obligation towards continuing education during the professional occupation. The majority of companies have a lack of trained employees. They need special qualifications on demand and normally do not have the time to wait for three years, like in a dual vocational training. The HRDC can fit the demand in special fields of engineering competences, like control and electric measurement. A company can apply via the Chamber of Industry to join the programme.

To enhance the curricula in terms of innovation in public and private universities, accompanied measures like vocational training or industry-related education could be an appropriate strategy to increase the awareness and the innovation capability of students. Corresponding elements can be implemented in the curricula without too much investments or efforts needed, provided that responsible Syrian public authorities are really open for this approach. Maybe more efforts are needed to advice the teachers how to train such new innovation related elements in practice.

[15] mentioned by the interview partner from the Ministry of Higher Education

Findings: Foresight R&D Agenda

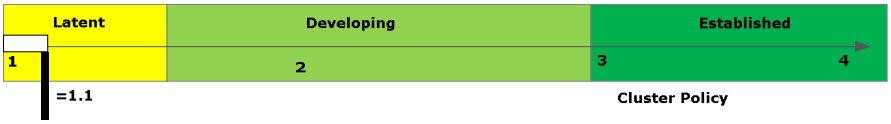


During the interviews, the questions regarding R&D foresight have been answered quite positively by the experts involved. In fact, a sustainable foresight-process to prepare future-oriented strategies is not established so far, but was ranked quite high compared to the other determinants.

There seems to be no clear responsibility for foresight-processes concerning R&D. The related issues are part in the long- and mid-term planning, but without a real strategic, demand oriented approach. The quite positive ranking of this determinant seems to be the result of the fact that many experts from academia and industry in Syria are sporadically involved in discussions about future technological developments and challenges, but not in a structured and politically supported way.

R&D foresight is an important political tool to identify future trends and potentials related to technology and innovation in a country. All partners which are related to R&D-activities have to be involved in such a strategic process, organised by public authorities being in charge on research and innovation. Actors out of the industry and the applied research sector also have to be involved. But, so far, it seems not to be an appropriate field of short term intervention in Syria now, since other fundamental determinants of innovation policy should be addressed first. The expected positive impact at that time seems to be quite low compared to other actions with higher impact.

Findings: Cluster Policy



According to the interviewed experts, currently there is no cluster policy or even items of it available in Syria.

Following the international discussions, clusters are one of the most promising tools to increase innovation and competitiveness on the regional level. Most of our experts are aware of the respective advantages and potentials. To most of them, it is obvious, that there is a strong need to focus on local branches with high potential and perspectives. A critical mass of actors on all levels is the mandatory nucleus for such a regional network or would-be cluster. There is actually no dedicated cluster policy in force. Nevertheless, we have noticed considerable interest from most interviewees in setting up regional networks and clusters. Most agreed that a dedicated national cluster policy may have a positive impact on Syria's innovation capability and competitiveness.

An assessment of promising sectors, analysing the strengths and weaknesses as well as the future opportunities should take place. After a following selection of the most competitive sectors (like renewable energy and water) the strategy defining and the implementation process starts.[16] Such actions are deducted by a national cluster policy. A national cluster policy can be defined and implemented with moderate efforts and investments. There are many good practices world-wide how nations with limited resources have successfully implemented clusters and regional networks.

4.2 Institutional Innovation Support Level

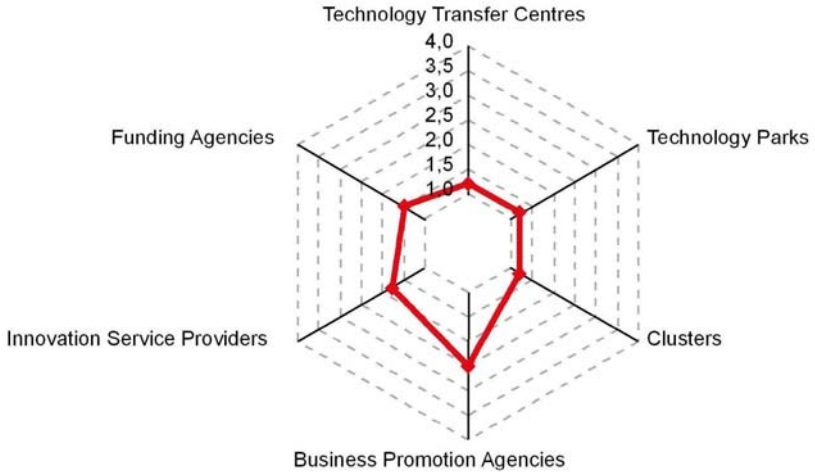
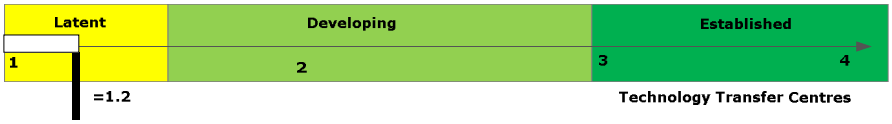


Figure 13 Indicator analysis institutional innovation support level - Syria

The six determinants of the institutional innovation support level provide a realistic overview of the interviewed expert of how far certain determinants related to institutional innovation support are developed. Most of the respective determinants are only latently developed, whereas business promotion agencies and the innovation service providers seem to be in a much better stage. Especially business promotion agencies were ranked top among all 29 determinants in Syria.

[16] Buhl, C; Meier zu Köcker, G.; (2008), Regionale Netzwerke initiieren und entwickeln, www.kompetenznetze.de

Findings: Technology Transfer Centres

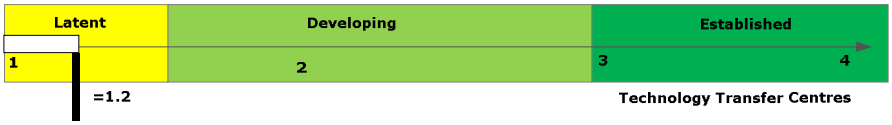


The interviewed experts agreed that in most cases neither the academic and research level nor the existing industry in Syria is actually in the position to develop new, competitive technologies or products according to the need of local and international markets. Nearly all technologies and sophisticated products have to be imported from abroad. Technology transfer in practice is considered to be underdeveloped, there are nearly no bridges between academia and the industrial sector. Technology centres are considered as an important tool to bridge the gap and to support the industry in developing products and technologies needed by the industry. As far as the experts are concerned, so far none is established in Syria.

To fulfil the future needs of the Syrian industry, first those needs must be identifiable. It is the impression by the interviewees, that due to a lack of communication the demand towards technologies and innovative services is presently unknown. Better synergies between research and industry are needed to enable the economy to reach more competitiveness and a stronger position in neighbouring markets. As soon as the technology transfer centres will be established, the responsibility should be placed either on the ministry level or by institutions like the Higher Commission for Scientific Research. Besides technological expertise the interviewees also see other innovation supporting activities, like pilot projects or trainings (innovation management or technology development) to be offered by such centres. There are plenty of good practises, e. g. in Egypt, where technology transfer centres gained an important role in offering demand oriented support in terms of innovation and technologies towards the local industry.

The formation of technology transfer capabilities could be one of the most relevant steps to improve the innovation capabilities of companies within the Syrian NIS, which combines the supply and demand side of technology knowledge and related facilities. The mandatory efforts connected by the setting up of such centres are considered to be very high in terms of financial investments in building, equipment and experience staff, but with comparably high positive impacts on the NIS.

Findings: Technology Parks



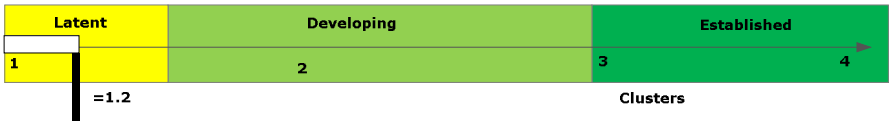
At the moment there are no technology parks established in Syria. The successful realisation of the idea of technology parks needs a fundament of innovative companies, applied R&D-institutions and a network of support.

Technopolis or Technopols as successful models for clustering high-potential technologies and stakeholders are well-known and on the strategic time-frame of the Syrian government. All ideas are still in the status of a first draft. There is a number of competitive and innovation-related com-

panies as well as a entrepreneurial spirit in the country, which could really benefit from such technology parks and the services provided by them.

Technology parks are more likely an approach for the mid-term perspective, with high potential impact, but as well with quite high investments. It appears useful to set up other institutional innovation support institutions, less expensive pilot activities, like incubators or technology transfer centres.

Findings: Clusters



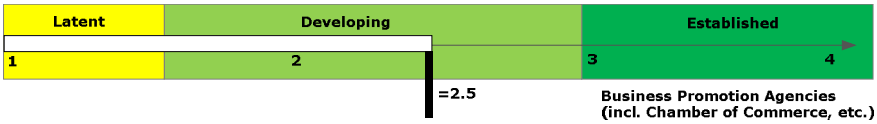
There are no clusters established so far in Syria so far. The existing industrial zones could be a future starting point since a certain kind of industrial agglomeration is given in such areas.

Four industrial zones (or industrial cities) are located in Syria: Adra in rural Damascus, Homs, Aleppo and Deir az-Zor. They offer business infrastructure and support special branches. For example, in cooperation with the Aleppo Chamber of Industry and the Syrian Enterprise and Business Centre SME (leather and footwear manufacturers) receive services and tools to assist them in achieving growth and increasing their competiveness. SEBC held a focus group with the aims to introduce the cluster concept and to encourage the manufacturers of the leather and footwear sector to form clusters, including benefits and added values which may arise from a cluster in this sector.

These kinds of examples could build up the idea of clustering and focused activities. There is a growing awareness of clusters and the potential benefits connected. If such clusters are capable to provide demand oriented services to its members, there are plenty of good practice examples world-wide, how firms have benefited. The main challenge, however, will be to gather respective companies, to analyse their demand and to motivate them to co-operate. The cluster management then must be able to set up specific services and added values, from which the cluster members can benefit from. The idea of setting up clusters could be promising in several branches like green tech, textile, ICT, life science, health care, mechatronics or food, where a critical mass of cluster members may be given. A regional and technological focus of such a cluster seems to be mandatory for future success.

The cluster approach could be established for one or two of the main industries in Syria as a test bed. As mentioned before, a deeper analysis of the economic and technological potential of certain regions, including the real industrial demands has to be done at the very beginning. The expected impact (specially the competitiveness advantage) could be very high, whereas the efforts and investments needed are quite limited. The main success factors will be the preliminary analyses and the cluster management which must motivate cluster members to cooperate and to create new ideas.

Findings: Business Promotion Agencies (incl. Chambers of Commerce, etc.)



The ranking of the agencies for business promotion is comparatively high, which leads to the conclusion that such kind of institutional innovation service provider is well established in Syria. It is obvious for all partners, that there is already a reasonable setting of corresponding infrastructure.

The existing mix of publicly funded and private business service providers offers reasonable services. Some of the interviewees would like to see a better coordination. For this target group, the Syrian Enterprise & Business Centre is an important and successful actor. Examples of publicly funded activities include the Syrian Investment Agency and the General Establishment of Free Zones. The Chambers of Commerce and Industry provide a set of services to their members (such as regular publications and bulletins and training courses).There is already a range of support activities towards enterprises, organised by the chambers, other NGO and financed by ministries, utilising international funding programmes.

For instance, within the framework of the tenth five-year-plan which aims at achieving sustained high growth rates, developing the industrial sector and increasing the role of the private sector, the industrial modernisation and upgrading programme (I'MUP-SYRIA) was adopted in 2007. Details about the programme, which is not related to innovation issues, can be found in ANNEX I.

Business Promotion is one of the most promising approaches to reach competitiveness in Syria (e. g. the framework of existing institutions needs better coordination and input of external know-how and a practical-based toolset). Further improvements would be connected with comparable high efforts needed and are therefore not in the focus of recommended improving actions, since this determinant is well developed.

Findings: Innovation Service Providers



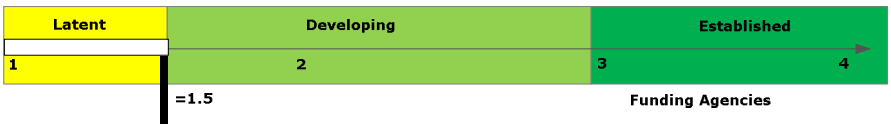
The phrase innovation is sometimes too general for the common understanding among the Syrian stakeholders. It seems that most of the interviewed partners had a diffuse understanding of the meaning and the relevance. More activities to improve the public understanding of innovation and technologies may be helpful. Many companies are not aware of the fact that innovations can provide better competitiveness, also tools, like innovation management, are highly appreciated by the actors of the Syrian NIS.

Enhanced innovation awareness and cultural behaviour would be a necessary precondition. Innovation related training measures should be designed with the special focus on topics with highest demand or towards selected sectors. So far, only a few institutions exist in Syria, acting as would-be innovation service providers. Nevertheless, this determinant is ranked comparably high to the

others. There is a need for technology (for example renewable energy) related training measures holistically transformed into activities.

Innovation support services and appropriate institutions are really needed in Syria. Demand oriented awareness campaigns and training measures could be appropriate to reach short-term effects. There are many options possible to proceed in this respect. The mandatory efforts seem to be quite low compared to the potential positive impact on the NIS. Strengthening innovation service providers and the spectrum of services they can offer to Syrian firms can be an option with a comparable high leverage effect.

Findings: Funding (and Implementation) Agencies



In general the administration of funding innovation and research belongs to the Ministries or the Higher Commission for Scientific Research; there is no dedicated responsibility beside the provider of grants. The practical implementation of funded project is considered to be a challenge.

Funding programmes for innovative projects or applied research are very rare in Syria. The Higher Commission for Scientific Research offers direct project funding. There are no agencies specialised towards the management of funding programmes. The Syrian Enterprise & Business Centre (SEBC) is one positive exception. SEBC is one of the major drivers for innovation related services, both for SME and for entrepreneurs who wants to implement new ideas into products and services. The permanent staff is a group of 40 well trained experts who offers services and trainings for the private sector. The self-conception of the SEBC is an implementation agency for projects aimed at developing the private sector in general and SME support & development in particular[17].

Funding and implementation agencies are very common in developed NIS where the governments are spending a significant amount of public money to science, technology and innovation. Such agencies are in charge of design, management and monitor public funding programmes. As long as the Syrian government does not decide to spend a significant amount of money for innovation support programmes, it does not seem to be appropriate to focus on the setting up of such agencies.

[17] www.sebcsyria.com

4.3 Programmatic Innovation Support Level

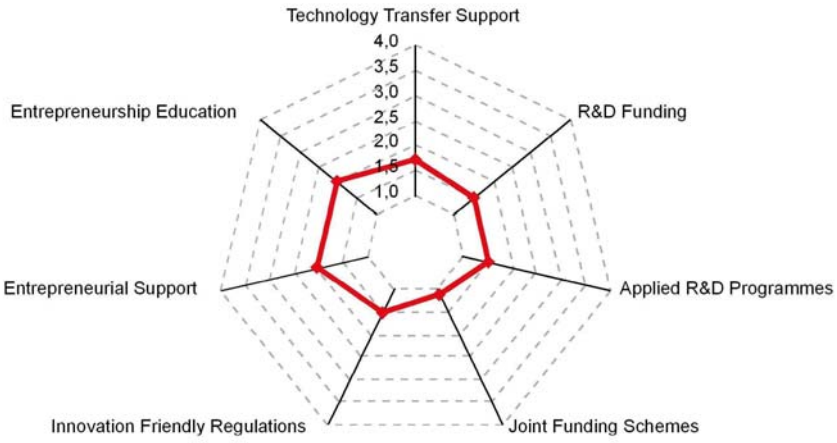
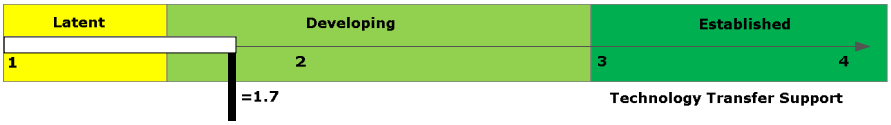


Figure 14 Indicator analysis programmatic innovation support level - Syria

Figure 14 reveals the seven different determinants related to the **programmatic innovation support level**. The pattern appears to be well balanced, most values lay between 1.5 and 2.0. This means that the most determinants are in the developing phase. The chance for current and future entrepreneurs to receive appropriate education and support are fairly high in Syria, since entrepreneurial support programmes are available in Syria. Other programmatic innovation support is not that far developed.

Findings: Technology Transfer Support

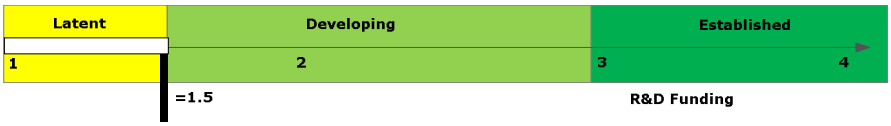


According to the interview partners, the technological infrastructure in Syria has room to be further developed in comparison to neighbouring countries. Technology transfer is weak whereas appropriate budget are very low compared to the national demand. This status is considered to be crucial by the interviewees, since technology transfer is seen as one of the most critical challenges for the Syrian economy.

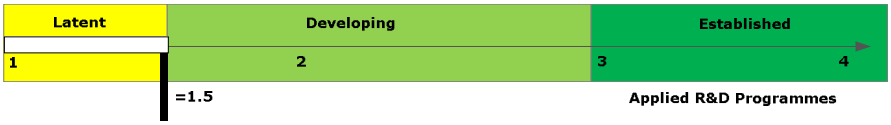
Syria's forthcoming membership in the WTO, the Syrian–EU association agreement and other agreements, are pressing the need to systematically address the productivity and competitiveness issues. Therefore, a number of programmes for enhancing the technology infrastructure in Syria, solving the technical problems faced by industrialists and deepening the understanding of the concept of innovation, are in force. This leads to an over-average value of this determinant. But dedicated support measures addressing this issues and technology transfer support programmes as well as appropriate public funds are far away from the national demand. The requirement for easier access to the limited programmes is also a feedback given by the interviewees.

Enhancing the relations between research and industry is one of the main challenges in Syria on the way to an innovation-driven economy. There are fairly good starting conditions, due to the fact that the demand for appropriate actions is already very high. There is a need for further development of culture, openness and supporting structures. Sustainable effects are not reachable in short-term. Nevertheless, any potential improvement of this determinant is directly linked to the readiness of the Syrian government to provide more public money for programmes that facilitates technology transfer between industry and academia. Besides the funds, demand oriented programme design features and an efficient management of such programmes are the main success factors. It is obvious, that such actions will cause certain financial and coordinating efforts.

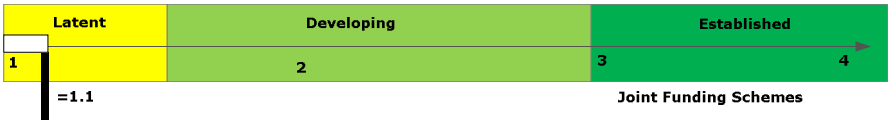
Findings: R&D Funding



Findings: Applied R&D Programmes



Findings: Joint Funding Schemes



The findings of these three determinants were grouped because the expert opinion was very similar.

Public funds for R&D are principally available in Syria (for example at the Higher Commission for Scientific Research). Funding schemes for R&D, for single or joint projects are generally available from HCSR. According to the EOS, joint projects are difficult, due to the uncertain situation of knowledge security. Environmental issues are some of the most important challenge for Syria and an opportunity for the industry, especially for SME, to join cooperation's with universities. Nevertheless, most experts agreed that so far there is a considerable gap between programmes and budgets available and the real need. Besides the limited funds, smart programme design features really address the national demand in terms of enhanced R&D. To match different competences form academia and industry is a highly appreciated approach by the interviewees.

For applied and even joint research the Higher Commission for Scientific Research (HCSR) offers funding with a budget of 250 million SyP (aprox. 390.000 €) a year for:

- Projects of universities, researchers and research institutions
- Support for diploma thesis for master and PhD studies
- Projects from SME

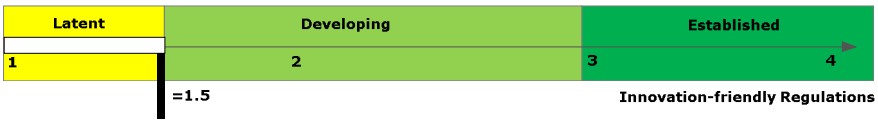
Applications for subsidies may be submitted directly at HCRS, forms are available on the website for interested parties.

The Higher Institute for Applied Sciences and Technology (HIAST) is a model, initialised 25 years ago, to conduct the excellence in applied research by selecting the best students from secondary schools and teaching them research skills in all applied sciences and technology fields. HIAST is following a French model, with international contacts (e.g. Germany and Russia). Students have to pay annual fees, which are partly sponsored by the government. HIAST provides suitable opportunities to make progress in applied research fields by joining courses to be awarded the degree of specialised diploma, master and doctorate.

Beside of its own strategy, the HIAST offers only little support for entrepreneurial ideas out of the institute. Relation of HIAST to the industry is close, but the industry is not very interested in results.

R&D-funding and joint-funding-schemes combine high expected impacts with high amounts of budgets. For this reason, these topics should not be high-ranked on the agenda of third party support unless the Syrian government is willing to increase its public investments. Successful funding schemes with long-term perspectives need a strong commitment of all involved parties.

Findings: Innovation-friendly Regulations



Although innovation is already a topic in the Syrian society, the enhancement of the innovative culture still needs public intervention. The responsibility so far is shared among many ministries and intermediaries without a clear structure of coordination. Regulations to enable and to support innovations are still missing.

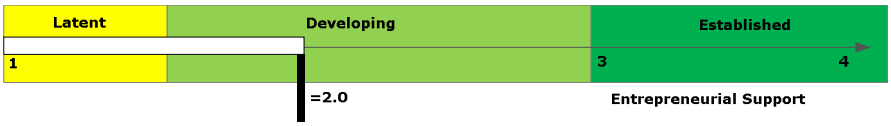
The political intention, to support innovation as a dynamic factor for the development of the Syrian economy, could be accompanied by more practical measures. Public companies, organised in general organisations (like for engineering), receive a certain amount of financing for the development of products and production infrastructure from the related ministries.

To convince innovators to transfer ideas to products and services, a clear policy of intellectual property rights is needed. The lack of a patent law to secure innovator rights and reliable institu-

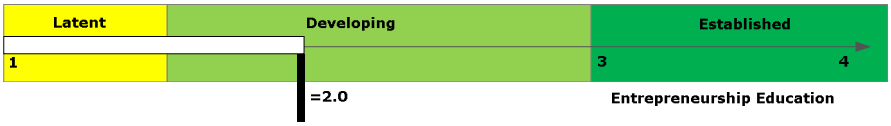
tions for support is one of the main challenges towards innovation-friendly regulations. Innovation management training is missing; there is the need to build the culture for innovation in advance.

It is a public task to build the breeding environment for innovation and competitiveness. Innovation-friendly regulations are one piece of the complete mosaic which is necessary, but more or less a result of activities and experiences in advance. There is less potential for the influence by external partners here, since this determinant is very much linked to national regulations.

Findings: Entrepreneurial Support



Findings: Entrepreneurship Education



The results out of the interviews for the two determinate regarding entrepreneurship support and entrepreneurship education are grouped, because the expert opinion was exactly identically to both determinate.

Entrepreneurship is a topic which was, according to the interviews, successfully revitalised during the last years in Syria. Support measures (mainly training and coaching) for entrepreneurship are available nowadays. Although the education in schools and universities does not motivate the ideas of self-employment, there is a growing interest among the younger generation. This seems to be a very important aspect. Still, awareness raising via media is a necessary precondition to further change attitudes in the future. Therefore it is well suited that Syria participate with own activities in the yearly Global Entrepreneurship Week.

There are several organisations working in the field of entrepreneurial support:

The Syrian Young Entrepreneurs Association is a non-profit civil institution that works to strengthen the entrepreneurial capabilities of young people in Syria in order to help them launch projects to make them job creators rather than job seekers. In order to do this, the association endeavours to deepen the culture of entrepreneurship among young Syrians, encouraging them and providing them with the necessary information and experience to launch and continue projects, in addition to

financing entrepreneurial activities. Since its launch in 2004, the SYEA has become one of the foremost authorities on entrepreneurship in Syria. The association has three branches in total: Aleppo, Deir az-Zor and its headquarters in Damascus[18]. The SYEA supports people between the age of 20 to 45 with trainings and consultancy. One of the main yearly events is an entrepreneurship competition with a price money of 1 million SYP (approx. 15.000 €).

The Syria Trust for Development is a non-governmental, non-profit organisation, established in Syria in 2007. Its aim is to empower people from all walks of life to play a full role in building and shaping their society. Syria's future depends on its people and the Syria Trust seeks to encourage individuals and communities to be positive agents of change.

In support of its project work the Trust also comprises four service groups – finance, HR, research and communications – which together ensure that the planning and business of the Trust is carried out to the best international standards and with proper accountability. The governance of the Syria Trust for Development is organised through a Board of Trustees chaired by Her Excellency Mrs Asma Al Assad.

The rural development division of the Trust emerges to implement work on the ground through FIRDOS, which is a project committed to improving the living conditions in rural areas by empowering individuals and communities to enhance their self reliance. FIRDOS operates in 60 villages in 6 governorates (Aleppo, Idleb, Lattakia, Homs, Hama, Quneitra).

Its activities and projects focus on several major areas: social development, economic development and institutional development.

The SHABAB Project was founded in 2005 as an initiative of the Syria Trust for Development. The Trust focuses on a few key areas of social development: education and learning, rural development and culture & heritage. The SHABAB Project is the flagship initiative of the Syria Trust's education and learning component. The other central project is the project MASSAR, which inspires cultural awareness among young Syrians under the age of 15.

The objectives of the SHABAB Project are:

- To boost entrepreneurship amongst young Syrians
- To encourage young Syrians to enter the business world
- To equip young Syrians with the key skills needed in order to succeed as productive individuals in society
- To create a positive impression of business amongst Syrian society

The mission of BIDAYA is to work with young people, by providing access to financial support to those with a viable business proposition but who are unable to find help elsewhere. The word BIDAYA means "beginning" in Arabic and is also an acronym for, "Boosting and Inspiring Dynamic

[18] www.syea.org/en

Youth Achievement." The programme will provide successful applicants with volunteer business mentors and full access to the organisation's local and national business support network.

The objectives of BIDAYA[19] are:

- To create jobs by offering an outlet to young Syrians with potential business force
- To contribute to the expansion of the private sector by deepening the pool of national entrepreneurs
- To help fill the gap of Syria's lending system by meeting the financial needs of very small businesses: needs that cannot be addressed by the conventional banking system
- To give hope and provide role models to unemployed young people
- To provide support and encouragement through mentoring and where appropriate, to provide guidance and training
- Special care should be made to finance environmentally sound businesses.

BIDAYA closed Memorandums of Understanding with all incubators in Syria to support mainly young entrepreneurs between the age of 18 to 35. It acts as an NGO since 2006 and offers low interest loans of up to 200.000 SYP (approx. 3.200€), business advisory for at least three years, opportunities for public relation support and a national and international network. BIDAYA is focused on grassroots projects. With a successful processing of the loans, graduates from BIDAYA have better chances to receive normal bank loans, because they can prove their credibility by clear credit documentation.

Determinants related to entrepreneurial support were ranked high, compared to others on this level. Many activities are ongoing and already designed under consideration of international examples and experiences. We noticed a positive impact as a consequence of previous actions. Nevertheless, since entrepreneurs are considered to act as key drivers for innovation, the number of entrepreneurs in Syria should be increased. There is a need for more awareness campaigns in all universities, research institutes and industrial regions in the country in order to attract more entrepreneurs. Further support measures are considered to be essential to further improve the entrepreneurial surrounding. International support in training and tool-transfer could be managed with less budgets and high potential impact.

[19] www.bidaya.org.sy

4.4 Innovation Capacity Level

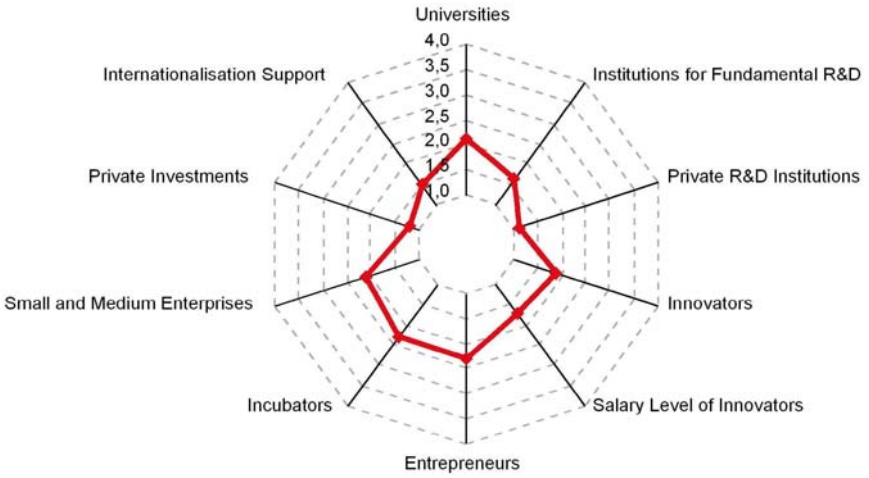


Figure 15 Indicator analysis innovation capacity level - Syria

The ten determinants reflect the status of development of the main actors of a NIS. The pattern appears to be very different according to the experts interviewed. The innovation-related performance of actors like SME, entrepreneurs, innovators or incubators seems to be on a positive way. Obviously there are only a few capacities regarding fundamental R&D or private R&D institutions in Syria. Capacities of private investors as well as international support related to innovation seem to be underdeveloped.

Findings: Public and Private Universities



It was the opinion of the interviewed experts that the landscape of universities in Syria is sufficiently developed. Many universities are offering a broad spectrum of disciplines. There are state universities and a growing number of private universities.

According to most of the interviews, the curricula, especially of the state universities should focus more on the practical demand of the industry. Practical, innovation related issues are very seldom, only a few professors have industrial experiences of their own or close connections to the industry. It was a common opinion, that the industry is not really interested in cooperation with the academic institutions for different reasons. So far, communication barriers hamper a closer co-operation between two groups. Nevertheless, the universities do have the potential to become more innovative and can principally be considered as a value partner for the Syrian industry.

In practice, there are some positive examples for contract research by universities for the industry. Nevertheless, universities should be more open for cooperation and practical technology transfer.

The government tries to enhance innovation as one of the main topics for the universities (new concept of higher education regarding the future needs of the market), by modernising the curricula. New private universities are more aware of innovation topics and some of them already have direct contacts with partners from the industry. In principle, there must be a demand from the private sector for better qualified graduates.

Overall, there should be a growing potential for pilot projects to prove the possible cooperation between private universities and industry, organised and managed by a mediator in the initial phase (for example chamber in close cooperation with an international expert organisation).

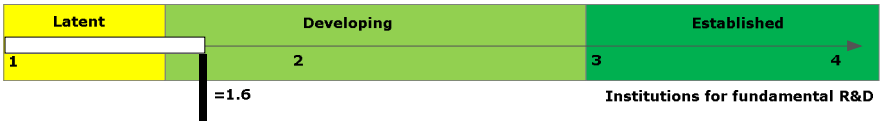
The acceptance of university degrees in society is still very high. Parents are spending a lot of effort to send their children to universities, independent from future perspectives after graduation. That should be a topic of awareness campaigns, because graduates are not really accepted in the industry. There is a common mistrust, the industry still believes, that universities are acting in an ivory tower.

The establishment of private universities, some are benefiting from professors from European universities, is a fairly new approach in Syria. The fee–based curricula (between 200.000 and 600.000 SyP) are open for all society levels that can effort this amount of money. At the moment, the majority of the private students come from Syria, others from neighbouring and Gulf countries. These universities try to benchmark their range of services with European partner universities and their standards. First graduates from private universities are in demand to find good jobs in Gulf countries, but most of them stay in Syria.

The awareness of self-employment-perspectives after the studies is not very common between students, neither form state nor from private universities. External experts could enhance the training capabilities of universities in the fields of innovation and entrepreneurship. The main goal should be the enhancement of entrepreneurship and innovation.

The respective determinants were assessed quite high, compared to others. Universities as such are traditionally not a field for external support. Universities in Syria could play a stronger role in educational issues, if they would adapt the existing curricula according to the industrial needs as well as by considering the industry as an important partner to jointly generate more innovations. Nevertheless, there are clear indicators that mainly the private universities are ready to become a stronger actor.

Findings: Institutions for fundamental R&D



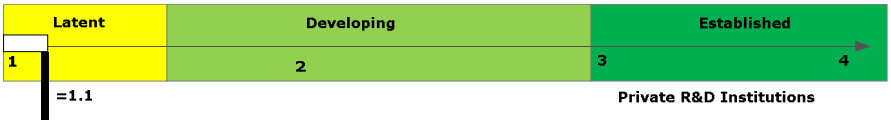
Institutions being involved in fundamental R&D are still quite seldom in Syria.

There are some institutions like the Industrial Testing and Research Centre or The National Energy Centre who are recognised as reliable partners for national companies and for the politic level (supporting the policy-making process by studies or evaluations). The budget of these institutes is covered by the Ministries. Management skills, infrastructure and staff need modern adjustments to

reach international competitiveness. To achieve better economic results, a direct, down-to-business-approach could be successful.

Fundamental R&D institutions are essential for a NIS to explore new paths with respect to science and technology. Nevertheless, such institutions heavily depend on public investments and programmes. So far, it is not recommended to focus on the improvement of this determinant, since it is strongly linked to others and significant financial and structural efforts are needed to be improved. On a long term run, universities can also take the responsibility for fundamental R&D in Syria.

Findings: Private R&D Institutions

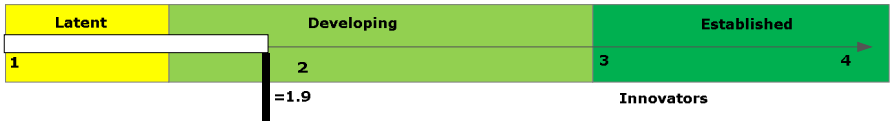


Private R&D institutions are almost nonexistent in Syria.

All product or services related research is part of the industrial companies. Private R&D institutions can only exist if significant private and public investments in R&D are available. However this is currently not given in Syria.

Similar to the Fundamental R&D, we currently do not recommend any specific action to this regards, since the framework condition for private R&D institutions is not given so far. This does not mean that there is no need for it. Private R&D institutions do play an important role in fully matured NIS. The financial and structural efforts, needed for setting up such institutions, including improved framework conditions, are huge with limited short and mid-term impacts.

Findings: Innovators



According to the interview partners, there is a lot of potentials for innovation and new ideas in Syria. Basically, engineers and scientists seem to be open minded and interested in to transferring there ideas into new products or services. Thus, good starting conditions are existing in Syria. That is the reason why the determinants were ranged comparably high.

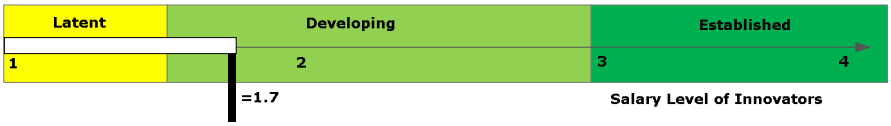
The unsolved challenge of intellectual property rights is considered to be a significant barrier, which keeps good ideas away from public perception. Once a year, the Syrian Inventor Association presents promising new ideas in the framework of a fair, open for public and private partners to contact the innovators directly and to initiate cooperation.

Nevertheless, following the interview partners, there is a lack of innovation in Syria. Related services to enhance the capabilities are needed. Through consulting processes, like during the

BIDAYA services, innovators become more realistic and demand oriented. These kinds of services should be directly addressed to young, creative graduates. Good examples of successful innovators are too little visible for the potential target groups.

To discover and to support innovators is an ongoing task for all institutions and actors in the NIS. As soon as first successful examples are visible in society, the culture of innovation will be revitalised and the number of innovators will rise.

Findings: Salary Level of Innovators

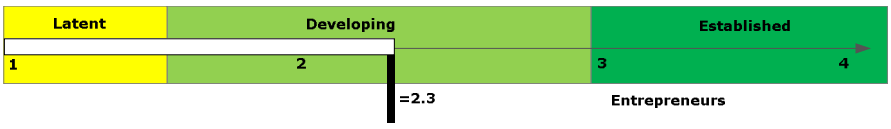


According to the EOS, the perspective for individual, secure income is much higher by transferring innovations into marketable products, than working in the public sector or being employed in a private company.

New ideas will be pushed forward only with a realistic horizon for private success and wealth for society. An adequate payment of innovators, compared with appropriate possibilities for commercialising is considered to be a good base for producing more innovations. Nevertheless, other determinants from all levels strongly influence the framework conditions innovators are operating in.

The compensation for innovation related activities is directly depending to the success of the invention in the market. There is no direct possibility for external intervention.

Findings: Entrepreneurs



Syria has a long tradition with entrepreneurial culture and trading capabilities in all levels of the society. In consequence of the economic policy of the last decades, self-employment as an option is anchored in society. Nevertheless, the fear towards an uncertain situation is still very common. Due to many public interventions in the recent past, these determinants were ranked quite high by the involved experts, especially compared to other determinants of this level.

Entrepreneurship is a topic in Syria, the population in average seems to be interested in being self-employed and in creating personal and society wealth with own businesses. The culture is in principle internalised, but it needs a well-designed set of measurements to reach sustainable success. For this reason, a range of support institutions with a framework of support measures is very active to revitalise the entrepreneurial culture in Syria. Although entrepreneurial education is part of the curricula only at private universities, there is a growing interest among students and graduates to develop themselves towards risk-taking entrepreneurs. Established companies and

even new established enterprises, have normally no or a very low strategic focus on new technologies. This can change into new opportunities for entrepreneurs.

According to the World Bank Country Profile "Doing Business 2009", Syria is ranked 124 overall for starting a business (figure 16):

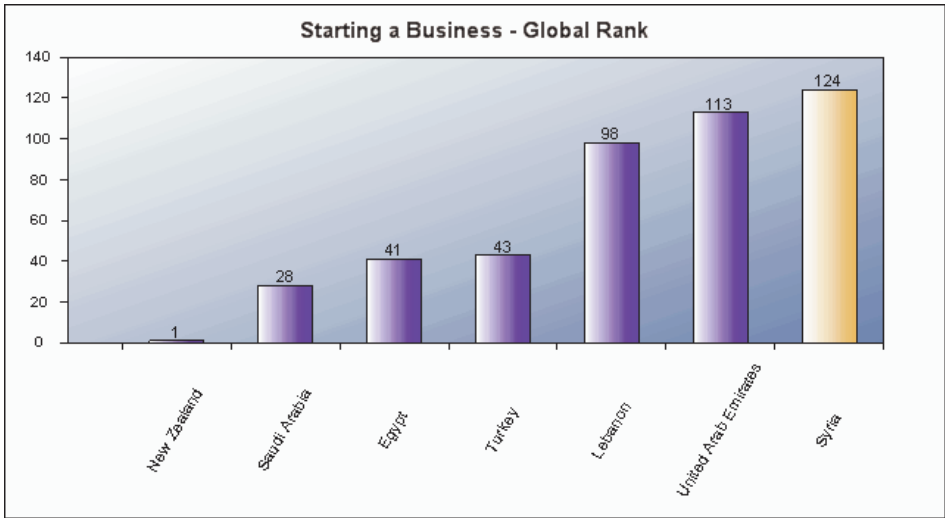


Figure 16 Global Rank for starting a business[20]

Main reasons for the weak position in the international context are mainly the administrative procedures and the duration of days for the registration of a new enterprise. The following figures prove the positive development of these indicators during the past years, benchmarked to the average of MENA countries:

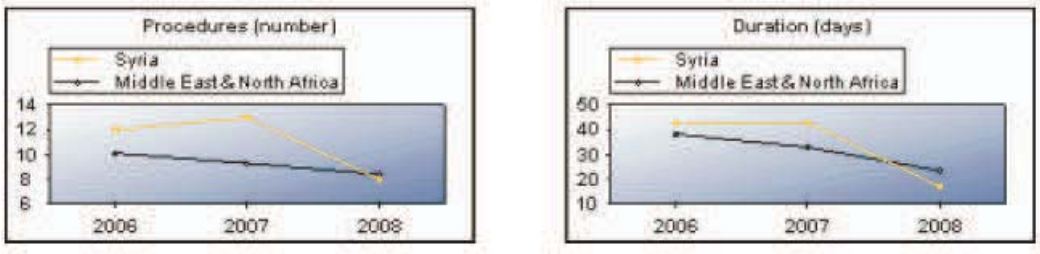


Figure 17 Development of selected indicators for starting a business

Access to finance sources seems to be one of the major problems for entrepreneurs in Syria. There is no risk capital available to finance the starting phase of innovative enterprises. The private capital market is very weak.

[20] World Bank Country Profile "Doing Business 2009"

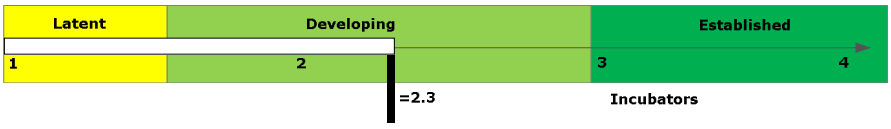
- Young people with sustainable ideas for new services and products could be forced by entrepreneurial trainings and an appropriate infrastructure (see also chapter “Entrepreneurial Support”). As one initial activity, the awareness of all related topics must be increased to attract the idea of self-employment.

As recommended above, it seems to be promising to further support the creation of new firms. Although much success was gained in the past, further support seems to be recommendable. The needed efforts are comparatively low; the expected impacts are high and reachable in a short time-frame.

Some possible activities were mentioned during the interviews:

- **Entrepreneurial conferences in the regions of Syria, in close cooperation with the universities**
- **Exhibition of good examples (local and international) of entrepreneurial business approaches – organised and supported by the patronage of European partners to enhance the credibility**
- **Courses for entrepreneurial skills at all technical universities**
- **Funding programme for entrepreneurs**
- **Business plan competition**

Findings: Incubators



Incubators are well recognised facilities for the improvement of entrepreneurship and technology transfer in Syria. Consequently, this determinant is ranked comparably high. The general idea has mainly been established by the first initiatives like the ICT Incubator which reached high international recognition. But still, the awareness among the community of innovation should be enhanced to achieve long lasting effects with high sustainability.

There are several incubators in operation in Syria, the first initiatives started in 2001. One of the nearly always mentioned initiatives is the pilot project “Information Technology Incubator” in Damascus, run by the Syrian Computer Society. A Technology Park is being planned by the Syrian Ministry of Telecommunications and Technology and will be located in Dimas, close to Damascus. An e-village will specialise in ICT and will include an incubator.

ICT Incubator - Syrian Computer Society

The Syrian Computer Society adopted the idea of establishing the first ICT Incubator in Syria, based on the conclusion of a number of studies. A study presented by (ESCWA) was taken into

consideration where the role of the Syrian Computer Society was to allocate the staff and the financial base.

The ICTI building was provided by Damascus University in return for symbolic fees as a sign of support for the incubator. The incubator is located within Damascus University's campus in Mazzeh, where incubated projects have the advantage of being around students and their activities.

One of the main goals of the Syrian Computer Society is supporting the Syrian ICT community in new economic and social projects, by encouraging the youth with their creative new ideas that add up into the labour market. The Syrian Computer Society established the ICTI in Damascus for this purpose and will be followed by similar incubators in the rest of municipalities around Syria.

Objectives:

- To affect policymaking
- To create jobs
- To develop profitable enterprises
- To provide income generating opportunities for disadvantaged populations
- To raise the awareness of potential entrepreneurs

Services of Pre-Incubation

- A short 4 to 6 months practical training
- Limited but free services
- Working on business plan
- Preparing for incubator admission

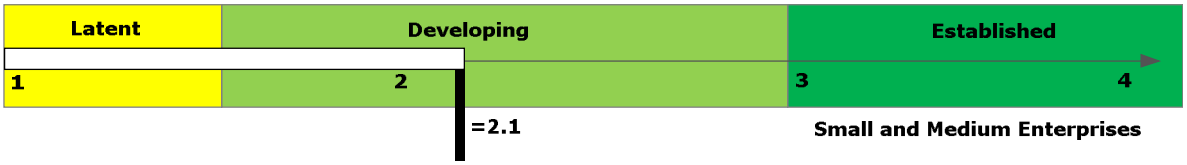
Services of Incubation: 18 months

- Full equipped office space
- Facilities
- Consulting (technical, legal, financial)
- Access to funding resources
- Business training

The Village Business Incubator (VBI) is a service centre for the development of female entrepreneurship in rural areas of the Lattakia Governorate. Based on the successful results gained during the first phase of the VBI project, the VBI was secured another milestone in April 2009 for three years. Five new villages are entitled to get the VBI services along with the villages targeted in the first phase.[21]

Although there are some promising activities with positive impact, setting up additional incubators could improve the framework conditions of entrepreneurs. On option could be offering of pre-incubation services at universities. The establishment of close relations to European incubators or associations to support incubation might be helpful for stimulate experience exchange and to increase the international reputation of the Syrian incubators. The corresponding efforts seem to be quite low, compared to the expected positive impact.

Findings: Small and Medium Enterprises (SME)



The overall number of private SME in Syria is estimated by official ratings with over 300.000, which produce more than 60% of the GDP. According the interviews, most of them are very small, like micro enterprises, according to the European definition with difficulties to find qualified employees. In the main opinion, SME are facing a lack of innovative products and services. In addition, the capability of innovation management is underdeveloped. For sure, innovation is not the only challenge SME currently face.

Although, several schemes provide financial and technical support as accompanying measures to enhance the innovation culture are available, SME traditionally have low budgets for improvements. Some EU funds are available, but nearly no national opportunities to receive financing.

The Syrian Enterprise and Business Centre (SEBC) is a private Syrian foundation that resulted from a 10-year partnership between the Syrian government and the European Union. The aim of this partnership was to support Syria's transition to a market economy by helping the Syrian private sector compete in international markets and by supporting the development of business institutions.[22]

[21] www.vbi-lattakia.org/

[22] www.sebcsyria.com

The main projects are:

- The SME Support Programme (SSP), which started 2006 and ends in February 2010, with a total EC funding of Euro 15 million €. With this funding, SEBC tries to achieve sustainability as a national institution providing a full range of services to the private sector and the government. The organisation offers services to the macro and the micro level. SME can receive technical assistance.

- The Trade Enhancement Programme (TEP), which started 2009 and will last till 2013, with a budget of 15 million €; the purpose of the TEP is to introduce trade related reforms that will progressively bring the Syrian trade related regulatory framework and practices in line with best international standards and to imbed administrative and legislative modalities, procedures and mechanisms necessary for trade liberalisation[23].

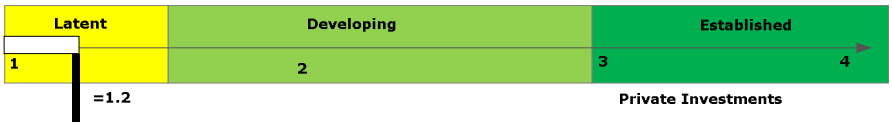
- The Business Environment Simplification Programme (BES) started as well in 2009 and will last till 2013. With a budget of 5 million € the project will assist the Syrian government in addressing issues of simplification of the business environment. The work carried out will complement the many studies that have already been produced by various bodies and will assist in the implementation of the 10th Five Year Plan (FYP) policies that promote the development of a 'social market economy'. The project will be attached to the Ministry of Economy and Trade which is responsible for the majority of business related legislation and regulation[24].

The strategic approach to focus on SME capabilities by establishing a SME agency could be a promising way to strengthen the position of SME within the Syrian NIS. Clear objectives and tasks as well as dedicated services for the SME would be the key success factors of such an agency.

[23] www.sebcsyria.com

[24] www.sebcsyria.com

Findings: Private Investments

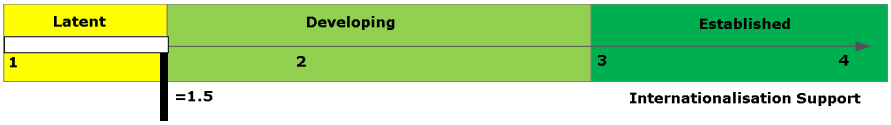


There is a financial shortage of financing or funding for innovation projects in Syria. Beside of public funding, national and third party sources, other private investment or risk capital is not available. Some programmes to make micro credits accessible were established recently. Consequently this determinant is ranked low compared to others.

Private banks were founded in 2004 and the liberalisation of the financial and insurance sectors has started slowly. A positive impact can be seen so far. An improvement of the regulatory and legislative structure to open the banking sector would be very promising. A good starting point would be the establishment of a credit guarantee scheme built as a partnership between the public and the private sector. Lately, there was nearly no foreign investment into the Syrian industry.

The development of the financial sector is outside the focus for possible external intervention, whereas the expected impact on the NIS would be very high.

Findings: Internationalisation Support



According to the interviews, the available support to redirect Syria onto the map of international recognition mainly comes from the European Union. The Medibtikar Euro Med Innovation and Technology Programme is one of the main donors.

Medibtikar[25] is derived from **MED**iterranean and **Ibtikar** (Arabic for Innovation). The Programme's budget is 7.3 million € and it will run till 31st December 2009. The beneficiary countries are: Algeria, Egypt, Israel, Jordan, Lebanon, Morocco, Syria, the Palestinian Authority, Tunisia and Turkey.

The beneficiary organisations are public and private enterprises concerned with increasing the competitiveness of SME. The Medibtikar project will provide the MEDA countries with new and improved instruments to stimulate innovation at the firm and country levels. The project will also simulate networking across the MEDA region and between the region and the countries of the European Union.

[25] www.medibtikar.eu/spip.php?lang=en

Objectives:

- To encourage good practice in **technology and knowledge transfer** by administrations, enterprises, industry federations, chambers of commerce, etc.
- To support the creation and/or improvement of **intermediary organisations** in charge of implementing support policies for SME (Innovation and Technology Centres (ITCs), TechnoParks, incubators)
- To support **innovation management**, from the development of a national innovation strategy to the identification of services to provide to SME
- To provide **support to specific sectors** facing common challenges in the MEDA region

"Strengthening Quality Management, Capabilities and Infrastructures in Syria" is an EU-funded programme to be implemented by a consortium lead by Belgium-based IBF International Consultancy on behalf of the Ministry of Industry and hosted by the Syrian Enterprise and Business Centre (SEBC). The 12 million Euro programme, with an implementation period of 4 years (2008 – 2012), will support Syria to raise its quality legislation and institutions to best international standards. This should result in enhanced quality of Syrian goods and services, better Syrian consumer protection and increased competitiveness of Syrian products in international markets[26].

The Chamber of Industry is in close cooperation with Italy and Japan, regarding the topics of industrial management and quality issues.

Syria should continue the strategy to intensify the relations to international donors and networks. The EU-connection is very strong, especially with Britain, France and Germany. As a full participant in the Euro-Mediterranean Partnership, Syria is in a very good strategic position to benefit from international opportunities.

[26] www.sebcsyria.com

5 Scope of Intervention in Syria

If a NIS should be improved, certain determinants must be addressed. It is quite clear that some determinants are easy to improve, others are much more complex. For this reason, we grouped the determinants according to their complexity for improvement. We used a portfolio with two different scales. One scale represents the “efforts needed” (How extensive is the needed amount of investment to enhance the performance of the determinant?), the other represents the “expected impact” (What range of improvement can be expected?).

To answer the question “What is the scope for intervention?” with highest perspectives for improvement of the capability of the NIS, the results of the interviews and the searched background information are visualised in the following portfolio:

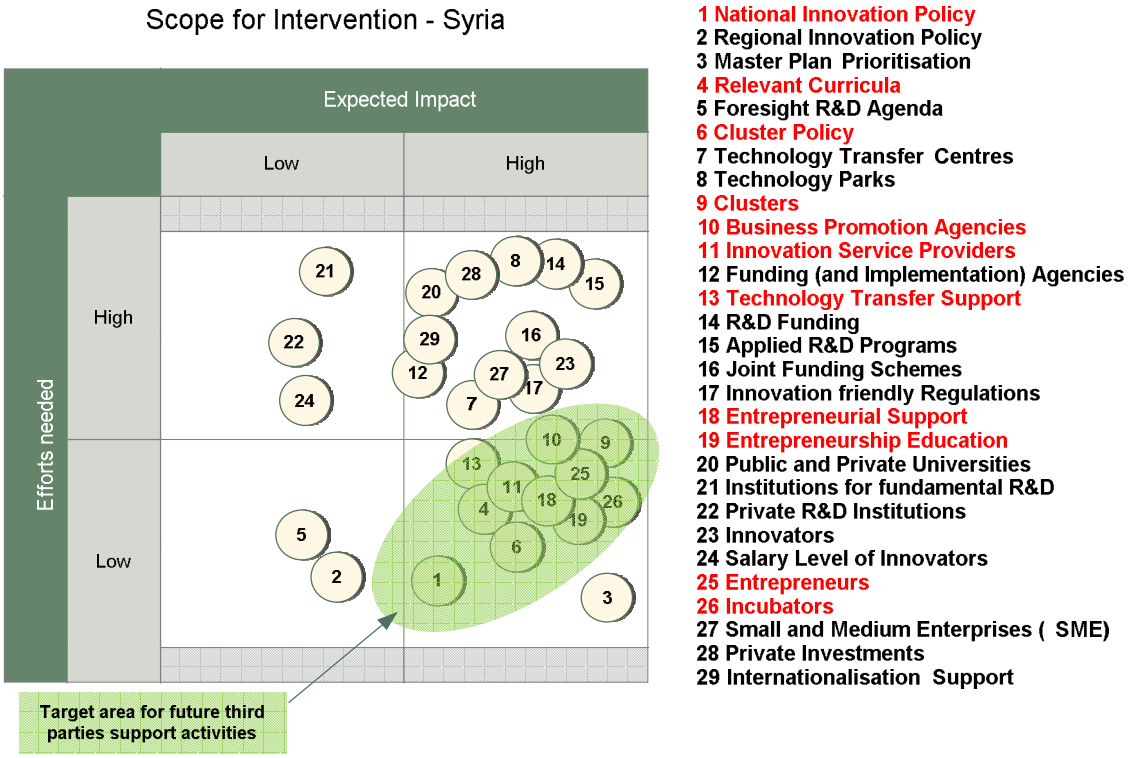


Figure 18 Portfolio – Scope of Intervention in Syria

The highlighted green area marks the most relevant and possible target area for future support activities to enhance the NIS. We are convinced that even with limited efforts significant improvements can be gained, provided the recommendations are consequently turned into practice.

The following recommendations describe first ideas for the initialisation and implementation of conceivable activities.

RECOMMENDATIONS FOR ACTION 1

Objective: The national innovation policy provides demand and practically oriented concepts and coordination measures to enhance innovation in Syria

Syria needs a nationwide guideline to establish the culture of innovation in all levels of the economy. There is a requirement for modernisation and streamlining of the service framework for the established industry, for new companies, for entrepreneurs and for potential innovators. The developed national innovation strategy should cover a policy-measure mix to reach midterm results, pointing out what the technological priorities for Syria are. Innovative policies must include a clear definition of the importance of modern R&D and state-of-the-art business behaviour for the competitiveness of Syria. It should be designed as a motivation for young researchers and students to take into account self-employment with the focus on innovative products and services. In result, the national innovation policy should build a proactive strategy involving a wide spread discussion process between all relevant stakeholders, related to academia, research, industry and the support network. Most important for the success will be the common understanding and the promotion of the policy. In the first phase of preparation the Ministry of Economy and Trade or the State Planning Commission should take leadership in the process. During the future process of implementation and renewing the strategy, more involvement of the intermediaries and support infrastructure will be crucial. New methods of funding could be an appropriate way to achieve strategic goals. There are good-practice examples from different NIS for successful funding programmes available.

The process to better implement a national strategy towards innovation as a national innovation strategy for Syria seems to be implementable with reasonable effort, but comparable high impact. Most of the necessary partners expressed their serious interest to play an active role and to be involved in the strategic process. High interest was also shown in terms of implementing such a national strategy. International support could assist the responsible bodies and actors with good-practice-examples, training and hands-on consultancy.

Recommended activities:

- Workshop for national policy makers
  - Good practice innovation strategies from Europe (Spain, France, Germany)
  - Benchmark with regional approaches
  - Transfer opportunities
- Setting up a high level expert group
  - Syrian representatives from policy (interministerial), academia and industry
  - Design a national innovation strategy
  - Some advice / impulse may be given by international experts
- Finding mission and study visit in Germany and other European countries

- Drafting the first innovation strategy
  - Feedback from international expert-network
- Set-up of a communication strategy

Main success factors:

- Strictly orientation towards the national demand
- National policy makers committed to long-term public investments in innovation
- National Policy makers with awareness of the needs of the innovation actors
- National Innovation Strategy focus' on competitive advantages of Syria
- Clear responsibility for implementation
- Regulatory and fiscal measures fully utilised for innovation

RECOMMENDATIONS FOR ACTION 2

Objective: Innovation support capacities are harmonised and strictly focused on strengthening the modernisation process of industry and research

To achieve sustainable results, all stakeholders in Syria with relation to technology, business development and innovation have to bundle their individual strengths to an efficient and powerful workforce. There already are excellent ongoing examples of support activities in Syria. Even the policy-framework is more and more related towards the modernisation process of the economy. Still missing is the coordinated approach to avoid inefficiency and misunderstandings. To provide the basis for business development, better synergy between industry and academia is needed. The synergy-process requires some kind of coordination, at the best offered by independent experts.

Business Promotion is one of the most promising approaches to reach competitiveness in Syria (e. g. the framework of existing institutions needs better coordination and input of external know-how and a practical-based toolset). Further improvements would be connected with comparable high efforts needed and are therefore not in the focus of recommended improving actions, since this determinant is well developed.

Enhancing the relations between research and industry is one of the main challenges in Syria on the way to an innovation-driven economy. There are fairly good starting conditions, due to the fact that the demand for appropriate actions is already very high. There is a need for further development of culture, openness and supporting structures. Sustainable effects are not reachable in short-term. Technology transfer support – the implementation of a stable bridge between industry and universities or research institutes is needed to enable absorptive capacities of technology and innovation. Good practice examples from Europe and neighbouring countries could help to build up a strategy with mid-term impact. Activities like Europe INNOVA or EUREKA are suitable benchmarks to learn from. These initiatives are well documented and offer good tools for know-how-transfer and for adoption of tools towards the Syrian circumstances.

Innovation support services and appropriate institutions are really needed in Syria. Demand oriented awareness campaigns and training measures could be appropriate to reach short-term effects. The already existent service providers in Syria could participate in special designed training measures, enforced by partners and with methods developed through practical experiences. Especially the business promotion agencies and the innovation service providers could benefit from these activities. All these different approaches could be combined together in one training unit with specified modules, related to the demand of industry and research.

There are many options possible to proceed in this respect. The mandatory efforts seem to be quite low compared to the potential positive impact on the NIS. Strengthening innovation service providers and the spectrum of services they can offer to Syrian firms can be an option with a comparable high leverage effect.

Recommended activities:

- Coordination by high level expert group
  - Setting-up of sub-groups
- Survey of existing nationwide support capacities – transparency
  - Avoiding duplications
- Identification of funding gaps
- Finding mission and study visit in Germany and other European countries
- Adjustment of existing measures

Main success factors:

- Involvement of the main actors in the Innovation System
- Willingness for transparency and cooperation
- Coordinated communication
- Break-up of traditional systems and responsibilities, if necessary

RECOMMENDATIONS FOR ACTION 3

Objective: The culture of entrepreneurship is acknowledged as one of the main elements to achieve economic development in Syria

Entrepreneurship is about creative people, their options and activities in starting, taking over or running a business. Entrepreneurship can occur in any sector and type of business. It applies to the self-employed and to firms of any size throughout the various stages of the business-life-cycle, from pre-start to growth, transfer and re-start. Entrepreneurship is relevant for firms in all sectors and technology-fields.

The promotion of entrepreneurial skills and willingness, to enhance individual's motivation and enthusiasm is a mission for public authorities. The long term impact for the society should build the driving factor for political activities. In Syria, all public activities should assemble a combination of the main intellectual and economical forces of the society. Entrepreneurship needs the influence and the interaction of research (universities), education (schools), networking (associations, e.g. of engineers) and business opportunities (private sector, national and international customer). The role of the state is visible in the responsibility for growth- and stability oriented strategies and macro-economics policies. The aims of such public activities are:

- Motivating and educating interested people to build up own enterprises
- Allocating the needed infrastructure, like commercial premises
- Enhancing the acceptance of innovation and the strove for independence

In the transition phase towards a social market economy in Syria there is a growing need for new enterprises and for the common behaviour of risk-taking among innovators. To build up the culture of entrepreneurship the already existing support activities could be enhanced by international knowledge-transfer and funding programmes offered by the Syrian government. The design and the implementation of these programmes could be supported by practically experienced partners from Europe. There are very successful support measures for all levels of entrepreneurial awareness and development in different European countries which could be used as practical benchmarks. Especially in the landscape of academia in Syria, entrepreneurial education could be extended. The skills among the teaching staff seem to be expandable by directly addressed activities, like train-the-trainer coaching.

Determinants related to entrepreneurial support were ranked high during the interviews. Many activities are ongoing and already designed under consideration of international examples and experiences. Nevertheless, since entrepreneurs are considered to act as key drivers for innovation, the number of entrepreneurs in Syria should be increased. There is a need for more awareness campaigns in all universities, research institutes and industrial regions in the country in order to attract more entrepreneurs. Further support measures are considered to be essential to further improve the entrepreneurial surrounding. International support in training and tool-transfer could be managed with less budgets and high potential impact.

Recommended activities:

- Conference “Entrepreneurship and Innovation Culture”
  - Lectures and practical workshops
  - Participants from Syria, neighbouring countries and Europe
- Coaching programme for innovation support agencies – “how to enhance the entrepreneurial culture and how to raise awareness among the young generation”
  - Participants: business promotion agencies, technology transfer and entrepreneurship supporter
- Training session “Entrepreneurial Education”
  - Practical excursions to European partners (like career services) and to establish (twinning approach) close relations with them
  - Participants: universities, technology transfer and entrepreneurship supporter
- Awareness campaigns in universities
- Adjustment of existing measures

Main success factors:

- Involvement of the main actors in the field of entrepreneurship support
- Willingness for transparency and cooperation
- Coordinated communication
- Openness for international cooperation and good-practice-learning

RECOMMENDATIONS FOR ACTION 4

Objective: Incubators and incubation services are established in Syria to enable the transition of innovative ideas into competitive products and services

Business incubators as the “infrastructural element” of an innovation System are a success story for many years all over the world and recently in several MENA countries. The approach of a secure environment for entrepreneurs, enriched with all needed services and networks still seems to be an enabler for business development and economic fortune. The idea has evolved over the time and has special characteristics depending on the local conditions. Strategies of branch-orientated incubators or women incubators are more common these days. The challenge for governments and development agencies is to develop capable and cost-effective solutions to create and finance incubator initiatives.

In many countries the process of structural reforms is still ongoing or at the beginning. To learn from good practice examples in the field of business incubation could be a helpful approach to avoid failures and time-lacks. The Syrian stakeholders could learn from the long-time experiences of European funding programmes in the field of business incubation.

Although there are some promising activities with positive impact in Syria, setting up additional incubators could improve the framework conditions of entrepreneurs. On option could be offering of pre-incubation services at universities. The establishment of close relations to European incubators or associations to support incubation might be helpful for stimulate experience exchange and to increase the international reputation of the Syrian incubators. The corresponding efforts seem to be quite low, compared to the expected positive impact.

To create or to redefine a promising incubation framework needs a deep investigation of the current situation and the future development of Syria. A partner with experiences and benchmarking-capabilities would be helpful to conduct a neutral analysis as a basis for strategic decisions.

Recommended activities:

- Deeper analysis of the current situation in Syria
  - Interviews and site visits
  - Recommendations for the strategy process
  - Participants: incubation management, innovation promotion agencies, technology transfer and entrepreneurship supporter
- Training concerning the design of business models for incubators
  - Recommendations for potential approaches to establish a sustainable business framework, mentioning environment strengths and weaknesses by an experienced European partner out of the practical incubation business
- Leadership-training for existing and future managers of incubation facilities
  - Supporting the involvement of Syria in the MENAinc Network (World Bank)

- Performance by: experienced Incubator Manager from European and the MENAinc Network
- Twinning-projects

RECOMMENDATIONS FOR ACTION 5

Objective: The innovation policy in Syria is focused towards cluster creation in future-driven sectors (like ICT or renewable energy)

The clustering of partners and activities to focus on selected branches with high potential for the regional and international competitiveness is one of the most prospective strategies for Syria. Cluster and networks are regional concentrations of interconnected universities, research institutions, service providers, small and large enterprises, educational institutions, public actors and further associated institutions in a particular field. They comprise the various steps of the value-added chain and are characterised by an intense interaction and communication among the involved players. Enterprises benefit from the research results in an early stage and likewise research institutions find business partners for implementing their research results more effectively.

Thus, clusters and networks are important as well as effective instruments for the promotion of innovation, development of economy and business as well as location marketing. Although there are no cluster initiatives in Syria so far, promising industry branches like renewable energy or ICT could build an excellent starting point for future activities. To start a cluster initiative the development of a suitable strategy would be the first step. This strategy should be based on an analysis of the capabilities, especially on the assessment of the institutional contexts. A deeper analysis of the economic and technological potential of certain regions, including the real industrial demands has to be done at the very beginning. The expected impact (specially the competitiveness advantage) could be very high, whereas the efforts and investments needed are quite limited. The main success factors will be the preliminary analyses and the cluster management which must motivate cluster members to cooperate and to create new ideas.

Recommended activities:

- Deeper analysis of the capabilities, especially on the assessment of the institutional contexts
  - Identification of would-be clusters (nucleus of clusters) or regional networks
- Design of strategy, technological focus and main services instruments
  - Recommendations for potential approaches to establish a sustainable cluster approach
  - Practical guide “how to build a cluster initiative”
- Leadership-training to build personal capacities for cluster management
  - Train-the-trainer approach - performance by: experienced cluster manager from Europe
  - Twinning-projects
- Implementation of cluster approaches
  - Coaching by external cluster experts

6 NIS of Syria, Jordan and Egypt - Comparative Overview

In the former chapters we conducted a comprehensive analysis of the 29 determinants as a status-quo assessment of the Syrian NIS. It was the main intention to characterise the determinants and to identify those that may be positively affected by third parties in the future.

This chapter is intended to present a comparative overview of the Syrian, Jordanian and Egyptian determinants. It is not our intension to rank these three NIS rather than to reveal country-specific strengths of certain determinants which may initiate a mutual learning among each other. Such benchmarking becomes more and more relevant in many fields. We selected Egypt for the benchmarking since its NIS is reported to be quite advanced compared to the most nations of the North African and Middle Eastern regions. Further more, we run an exploratory study on the Egyptian NIS in 2008 which gave us some value insight views[27]. Since the assessment of the Egyptian NIS was not in the focus of the Term of Reference, we were not able to run interviews with a comparable number of experts like in Syria. Thus, the values of the determinants do not have the same validity like for Syria, but we are convinced that they give a reliable picture.

Macro Level – Policy Level

The findings related to the determinants belonging to the macro level/policy level reveal that there seems to be no considerable difference between those three countries. The patterns are well balanced; the respective values are all around 1.5 and 2.0. This means the innovation related policies of all three countries seem to have comparable characteristics, or in other words, have the same strengths and weaknesses. In all countries there is no national, harmonised innovation policy available. Often some respective elements are regarded in certain plans and official documents. In addition, regional innovation policies or master plans for turning such policies into practice are not common. For all three countries, cluster policies do not really exist, although in Jordan some bottom-up clusters have emerged in the recent past. Finally, it seems to be worth mentioning, that the determinants for Jordan continuously ranked higher compared to the other countries.

[27] Meier zu Köcker, G; Hahn, P, The National Innovation System in Egypt – An exploratory study of Egyption Technologiy Transfer Centres, financed by GTZ, www.iit-berlin.de/veroeffentlichungen, 2008

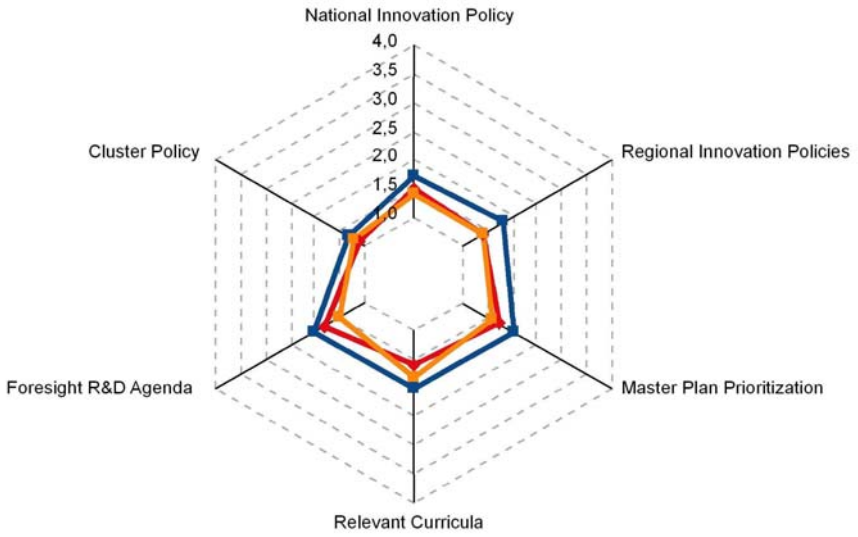


Figure 19 Comparison of the determinants related to the policy level (macro) between Syria (red), Jordan (blue) and Egypt (orange)

Meso Level – Institutional Innovation Service Providers

In the previous chapters we stated that institutional service providers are main tools to turn innovation policy into practice and often the main source to support the actors of a NIS in innovation matters. Thus, the characteristics of the respective determinants are quite important and reveal how innovation policy or official commitments are implemented in reality.

The determinants for Egypt are continuously ranked higher as for Syria and Jordan, except for clusters. The latter one does not really exist in Egypt, whereas in Jordan some clusters do exist. On the other hand, technology transfer centres are much more developed and receive stronger public support in Egypt compared to the other countries. The same can be seen for funding agencies and innovation service providers. The latter ones do receive better public support and provide demand-oriented services. Nevertheless, there is room for improvements even for the Egyptian institutional innovation service providers. As far as business promotion agencies are concerned, determinants of all countries reach comparable values. Although there a more technology parks existing in Egypt than in the other countries, this determinant was not ranked very high by the Egyptian interviewees since they often lack to provide real added values. Smart Village Egypt is reported to be a very positive exception. In some cases there are only buildings and no services are provided for the tenants.

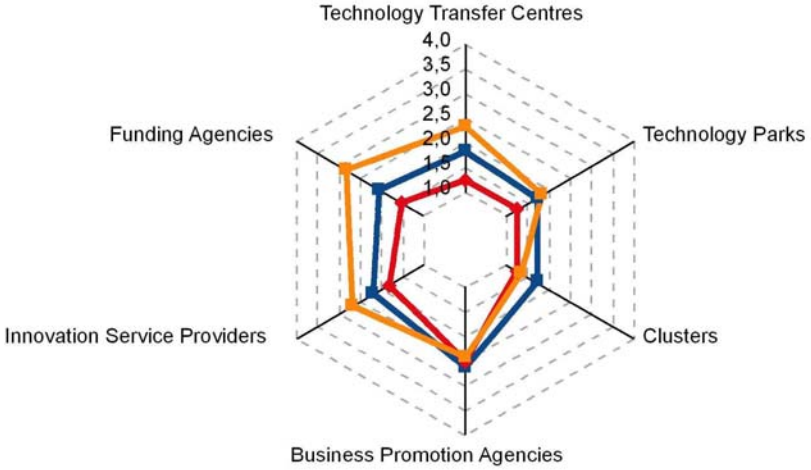


Figure 20 Comparison of the determinants related to the institutional innovation support level (meso) between Syria (red), Jordan (blue) and Egypt (orange)

Meso Level – Programmatic Innovation Service Providers

Programmatic innovation service providers are the second backbone of a national innovation support system. The patterns of the three countries are quite comparable. All three are well balanced with maxima on entrepreneurial support. The determinants for public funded programmes (R&D funding, applied R&D funding and joint funding schemes) are ranked lower, compared to the others. Here similar low values are reached, since the governments hesitate to significantly invest in public programmes supporting science, research and innovation. Nevertheless, Egypt invests much more in technology transfer schemes, not related to R&D. Entrepreneurial support is of high relevance in all countries, whereas the determinants of Egypt are ranked very high, compared to the others. This seems to be a clear outcome of significant efforts the Egyptian government has been spending for several years. Finally, we very surprised about the finding that the Syrian determinants are often close to those of Jordan.

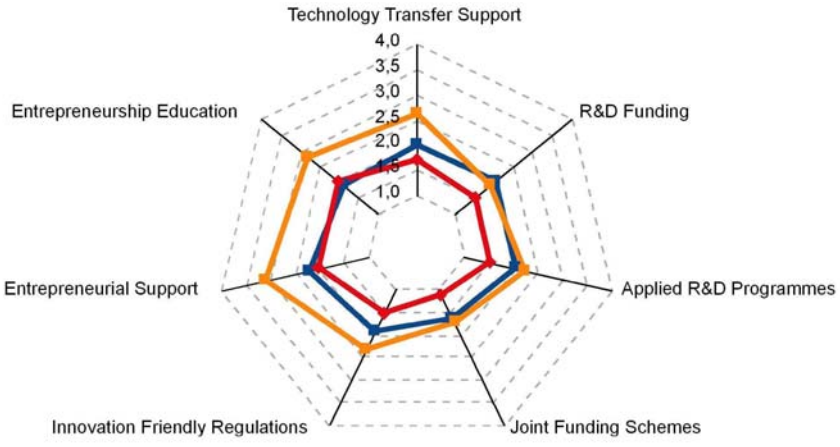


Figure 21 Comparison of the determinants related to the programmatic innovation support level (meso) between Syria (red), Jordan (blue) and Egypt (orange)

Micro Level – Innovation Capacity

The innovation actors of a NIS are the main drivers for innovation, strongly depending on the framework conditions provided by the respective governments. Figure 22 reveals the respective patterns. All three countries have different actors who have been ranked higher than in other countries. As far as Syria is concerned, innovators, entrepreneurs as well SME are assessed to be on the same level as in Egypt and Jordan. Universities and international support actions are considered to be most developed in Jordan. Private R&D institutions and investors seem to have the best conditions in Egypt, whereas the absolute values are not that high.

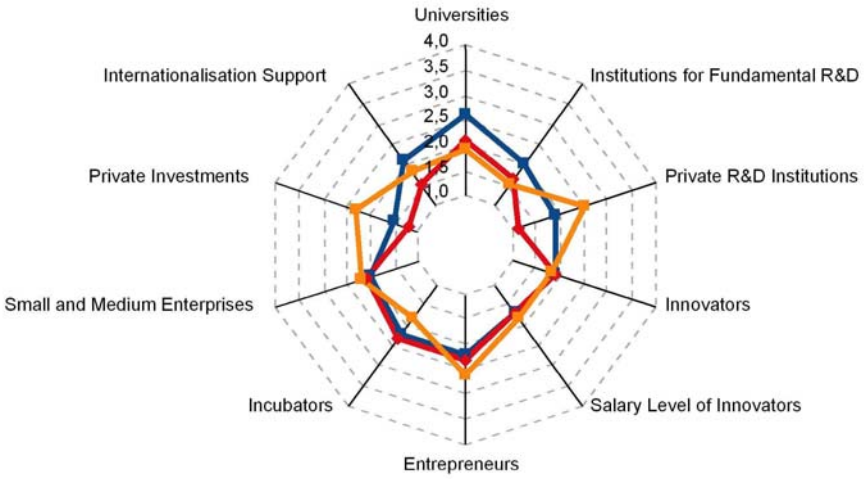


Figure 22 Comparison of the determinants related to the innovation capacity level (micro) between Syria (red), Jordan (blue) and Egypt (orange)

According to our perspective the findings clearly point out where specific countries have individual strengths and weaknesses. If policy makers or interested parties are keen to learn from good practice, there is no need to look always at Europe rather then to learn from the neighbours. For sure, in some areas, like public programme-based support initiatives, all three countries should gain experiences from other countries.

7 Appendix I: Programme-Descriptions and Other Related Documents

The Technology and Innovation Programme

Objective: Support applied R&D of relevance to the needs of manufacturing industries — **Action No: Technology 1**

Action : Implement a programme for applied R&D initiated by enterprises
Description / rationale
Missing linkages between R & D in Research Centres and Universities and the actual down to earth needs of industrialists is a problem in Syria just as it exists in a number of other LDCs. This action aims to initiate a pilot project whereby the gap between the two sides is narrowed. IMCS will finance through its R&D fund (suggested in policies programme) 80% of the cost of a number of research projects suggested by the private sector.

Status:

To be first experienced for textiles at the level of UNIDO pilot project to be later adjusted as needed on the basis of actual experience.

Responsibility:

IMCS,

Ministry of Higher Education,

Chambers/ Private sector

Implementation Steps

1. Through the public relations department invite industrialists (individually or collectively) to submit proposals for applied industrial research they need to enhance their production lines or solve specific technical problems they face.
2. Proposals are to be assessed by experts appointed by IMCS on the basis of clear criteria such as (sound scientific grounds, degree of usefulness for the sub-sector in question as a whole among others).
3. Viable proposals will be financed by R&D fund (80% to be paid by IMCS while the remaining 20% are covered by the enterprise).

Objective: Establish a network of specialised Technology Centres **Action No:**

Technology 2

Action : Establish three Technology Centres as a pilot project
Description / rationale
Technology centres are meant to provide specialised technical services to industrialists in the areas of Technology of Production and Innovation. This action is to initiate the network of Technology Centres with the establishment of three centres in textiles, food and chemicals as a first phase. The new centres will build on the existing infrastructure as much as possible. Depending on the special status of each sub-sector the new centre can range from simple up-grading of an already existing institution to the creation of a brand new centre.

Status:

New project but technology is receiving a lot of attention at the government level.

Responsibility:

Ministry of Industry,

Ministry of Higher Education,

Chambers/ Private sector

Implementation Steps

1. Appoint experts to identify the special needs of the chosen sub-sectors on the basis of through background studies (studies programme) and close interaction with stakeholders.
2. A scanning of the already existing institutions providing services to industrialists is to follow. This is to lead to a clear delineation of the role of the new potential Technology Centre and its most suitable structure.
3. Explore possibilities of twinning/linking the new centres with specialised best practice western Technology Centre to ensure future sustainability of high quality service delivery to industrialists within the sector.

Objective: Promote the concept of Innovation **Action No:**

Technology 3

Action : Arrange a series of workshops to promote the concept of Innovation

Description / rationale

Although innovation is often seen as an advanced step in a country where the technological infrastructure is weak and enterprises are generally still small and immature, it is important to educate the business community on the meaning and importance of the concept which is often understood only in its very narrow sense of new products. The reason lies in the fact that Syria is already too far behind and needs to make big jumps at all fronts.

Status:

New project.

Responsibility:

Ministry of Industry,

Ministry of Higher Education,

Chambers/ Private sector,

International consultants

Implementation Steps

1. Appoint international experts to conduct a series of workshops to explain the innovation concept and promote its implementation.
2. Workshops are to be attended by private sector producers and members from the research community.

SHABAB

The SHABAB Project was established to help a generation meet the new demands of Syria's labor market. The first programme to be piloted by SHABAB was the "**Business Awareness Programme**," which brings volunteers from the business world to share their knowledge and expertise with secondary school students over the course of a two-day programme. The programme is based on partnerships with businesses that provide not only financial support, but also donate their time and knowledge to the programme.

SHABAB now conducts five programmes (the others are described below) and now has over 41,000 direct beneficiaries nationwide. SHABAB partners with the government, the private sector and civil society in order to implement its programmes at a local level. Programmes have been implemented programmes in Damascus, Aleppo, Homs, Lattakia, Deir az-Zor, Rural Damascus and Quneitra.

In addition to the Business Awareness programme, SHABAB conducts five other programmes targeting 15 to 29 year olds:

- **Business Experience programme:** This programme places young people (15 to 18 years old) in a real business environment, giving them the chance to experience the business world from a first-hand perspective while further developing key, career-building skills.

- **Know About Business programme:** The "Know About Business" programme is comprised of a set of training materials to boost the entrepreneurial skills of young people and teach them how to start their own businesses. The programme was adopted as an official part of Syrian school curricula and is now delivered by trained teachers in vocational schools and intermediate institutes, with a more sophisticated version at Universities.

- **Business SHABAB programme:** The Business SHABAB programme is "an audio and visual extravaganza" aimed at convincing young people of the possibility of starting their own business. The programme engages with young people through music, sound bites and innovative media tools.

- **Business Clinic programme:** This programme offers young people primarily aged between 18 and 24 the opportunity to discuss their future career with an advisor. For those seeking employment, it provides them with job-searching skills whilst for those that are more entrepreneurial, it offers guidance on how to start a business.

Associations

National:

Syrian Computer Society

The SCS was founded in 1989 as a joint initiative by founding members who are specialised in the field of informatics. As a mediator between academia and ICT-Industry, the SCS has grown up rapidly. Actually it encompasses around 5000 active members and more than 7000 associate members, who come from different regions of Syria.

The SCS aims at introducing IT into various Syrian economic. It also aims to diffuse informatics culture by means of organising conferences, symposia, seminars, lectures and exhibitions, in addition to producing TV programmes and issuing pamphlets concerning IT.

The SCS aims to introduce and establish the ICT culture in the main economic sectors in Syria through the following activities:

- Organising conferences, symposia and seminars and exhibitions
- Developing a national programme for IT diffusion
- Developing a national academic programme
- Organising training courses on IT within the SCS's training and qualification activities frame-work
- Issuing publications relevant to IT.
- Providing incubation facilities for the youth, specially from the academic environment

General Organization of Engineering Industries

One of eight:

- General organization for engineering industries
- General organization for chemical industries
- General organization for food industries
- General organization for textile industries
- General organization for cement and building materials
- General organization for cotton ginning and marketing
- General organization for sugar
- General organization of tobacco

… is the organisation which belongs to the Ministry of Industry and is responsible for the engineer-ing public sector companies. 13 companies (with more than 6.000 employees) are affiliated to the

organisation. They are active in the fields of steel, gas, metal construction, manufacturing, fibre optic cables and renewable energy. The aim of the GOEI is the supervision of the companies and the lobby work for the branch. The association offers budgets for suffering companies to overcome short- or mid-term difficulties. The organisation receives fees, like a profit-share from the companies and a basic budget from the Ministry of Finance.

International:

UNDP (United Nations Development Programme) in Syria

The development objectives of UNDP Syria are:

- Strengthening and better targeting social protection
- Improving structures and climate for trade, investment and competitiveness
- Improving employment environment and opportunities for skill-enhancement for the under- and unemployed, especially women and youth
- Enhancing poverty alleviation programme including income generation and improving access to extension services in rural and poor areas.
- Policy and decision-making supported by quality information and analyses.
- Accountability of executive bodies reinforced, toward the general public and in regard to committed UN conventions.
- Democratic electoral processes and civic education enhanced.
- An empowered civil society involved in the development and implementation of public policies, planning and programmes.
- Improving administrative services for citizens and courts' administrations taking into account citizens' rights and the needs of vulnerable groups.
- Planning and decentralisation policies and structures enhanced
- Strengthened institutional mechanisms and policies for improving the legal status of women and increasing gender equality.
- National capacity strengthened for meeting obligations towards ratified environmental conventions (biodiversity, climate change and desertification conventions; and the Stockholm Convention on Persistent Organic Pollutants – POPs) and national environmental legislation enforced with a particular focus on water policies.
- Improved environmental situation with the involvement of local communities and the private sector
- Comprehensive and coordinated gender-sensitive disaster management system in place, with national and local capacity enhanced.

8 Appendix II: Literature

General Information

1

European Commission (Directorate-General for Enterprise and Industry); Organisation for Economic Cooperation and Development (Private Sector Development Division); European Training Foundation: Report on the implementation of the Euro-Mediterranean Charter for Enterprise, 2008 enterprise policy assessment, in consultation with: European Investment Bank; © European Communities/OECD, 2008

2

World Economic Forum within the framework of the Global Competitiveness Network: The Global Competitiveness Report 2009-2010, © 2009 World Economic Forum

3

Niels Bosma, Zoltan J. Acs, Erkko Autio, Alicia Coduras, Jonathan Levie: Global Entrepreneurship Monitor, 2008 - Executive Report, 2009

Syria

4

Prof. Lena Tsipouri, Prof.Dr.Eng. Samir Elsayyad, Dr.Eng. Ahmed Elsayad, Diagnostic Study of Syrian Innovation System, EuroMed Innovation and Technology Programme

5

European Neighbouring and Partnership Instrument; Syrian Arab Republic; Strategy Paper 2007 - 2013 & National Indicative Programme 2007 – 2010, EU-Report

6

INDUSTRIAL MODERNIZATION AND UPGRADING PROGRAMME IN SYRIA (IMUPS), Executive programme - Final report, UNIDO, 2007

7

Dr. Constantin Zaman: Institutional & Sector Modernisation Facility; How efficient is the Syrian Industry?, ISMF, 2007

8

Euro-Med Charter for Enterprise Initiative: Euro-Mediterranean Charter for Enterprise; Country Report; Ministry of Industry; Syria, 2006

9

United Nations Development Programme; The Economic Status of Syria; A focus on Women's Economic Participation;

10

World Bank and the International Finance Corporation: Doing Business 2009, Country Profile for Syria, © 2008 The International Bank for Reconstruction and Development / The World Bank

11

Consultative Group to Assist the Poor: Policy and Regulatory Framework for Microfinance in Syria, 2008

12

Syrian Arab Republic State Planning Commission: 10th FYP (2006-2010); The Five Year Plan Approach

9 Appendix III: List of Interview Partners

Syria

Institution
Ministry of Higher Education
State Planning Commission
Public Corporation for Employment & Enterprise Development
BIDAYA (Boosting and Inspiring Dynamic Youth Achievement)
Syrian Enterprise and Business Centre
Faculty of Mechanical & Electrical Engineering
Syrian Inventors Association
Human Resources Development Training Centre
University of Kalamoon
General Organization for Engineering Industries
Damascus Chamber of Industry
SHABAB
Syrian Computer Society – ICT Incubator
ICT Incubator
Ministry of Industry
Ministry of Economy and Trade
National Energy Research Centre
Higher Institute for Applied Sciences and Technology
CHAOUI & Co
Higher Commission for Scientific Research
Arab. Federation for Engineering Industries

Institution:	Ministry of Higher Education
Name:	Prof. Mohamed Najib Abdul Wahed
Position:	Deputy Minister of Higher Education
Date of Interview:	25th of July 2009

Institution:	State Planning Commission, SPC (GTZ)
Name:	Mr. Bassam Al-Attar
Position:	Director of Cooperation with Europe
Date of Interview:	26th of July 2009

Institution:	Public Corporation for Employment & Enterprise Development (PCEED)
Name:	ENG: Obeid Al Salem
Position:	Deputy General Manager
Date of Interview:	26th of July 2009

Institution:	BIDAYA (GTZ)
Name:	Mrs. Rana Shanawani
Position:	Chief Executive Officer
Date of Interview:	26th of July 2009

Institution:	Syrian Enterprise and Business Centre (GTZ)
Name:	Dr. Hisham Khayat
Position:	Senior Business Development Consultant SEBC Business Incubator Manager
Date of Interview:	26th of July 2009

Institution:	Faculty of Mechanical & Electrical Engineering (GTZ)
Name:	Prof. Hashem Oirkozek
Position:	Dean of faculty of Mechanical & Electrical Engineering
Date of Interview:	27th of July 2009

Institution:	Syrian Inventor Association
Name:	Mr. Adel Mohanna
Date of Interview:	27th of July 2009

Institution:	Human Ressources Development Training Centre
Name:	Dr. Reinhard Edelmann
Date of Interview:	27th of July 2009

Institution:	University of Kalamoon
Name:	Dr. Said Younes
Position:	Vice President for Academic Affairs
Date of Interview:	27th of July 2009

Institution:	General Organization for Engineering Industries
Name:	Eng. Nasser Al-Sheikh Youseph
Position:	General Manager
Date of Interview:	28th of July 2009

Institution:	Damascus Chamber of Industry
Name:	Eng. Emad Ghreiwati
Position:	President of Federation of Syrian Chambers of Industry
Date of Interview:	28th of July 2009

Institution:	SHABAB
Name:	Mr. Anas Dharweesh
Position:	Programme Development Manager
Date of Interview:	28th of July 2009

Institution:	Syrian Computer Society – ICT Incubator (VDI)
Name:	Dr. Hasna Askhita
Position:	Board of Directors
Date of Interview:	28th of July 2009

Institution:	ICT Incubator
Name:	Mrs. Rima Shaaban
Position:	Director
Date of Interview:	28th of July 2009

Institution:	Ministry of Industry
Name:	Mr. Wolfgang Weber
Position:	Advisor
Date of Interview:	29th of July 2009

Institution:	Ministry of Industry
Name:	Eng. Imadeddin Rjlh
Position:	International Relation
Date of Interview:	29th of July 2009

Institution:	Ministry of Economy and Trade
Name:	Mrs. Sulafah Radwan Akili
Position:	Director of SME's Directorate
Date of Interview:	29th of July 2009

Institution:	Ministry of Economy and Trade
Name:	Mr. Yasser Saada
Position:	Director of AL-Bassel Fair for Inventions & Innovation
Date of Interview:	29th of July 2009

Institution:	National Energy Research Centre
Name:	Eng. Mohammed Khalil Sheki
Position:	General Director
Date of Interview:	29th of July 2009

Institution:	Higher Institute for Applied Sciences and Technology
Name:	Dr. Wael Khansa
Position:	Director
Date of Interview:	30th of July 2009

Institution:	CHAOUI & Co
Name:	Naji Chaoui
Position:	Managing Director
Date of Interview:	30th of July 2009

Institution:	Higher Commission for Scientific Research
Name:	Prof. Ghassan Assi
Position:	Director General
Date of Interview:	30th of July 2009

Institution:	Arab. Federation & Engineering Industries
Name:	Dr. A. Haytham El Yafi
Position:	Director
Date of Interview:	30th of July 2009